

Computer vision tasks

Image segmentation & Object detection

Amit Shmidov | Tutorial 8

Relies on the corresponding presentations from CS231 2026, Stanford university

Computer vision tasks

Image classification



Cat

No spatial context

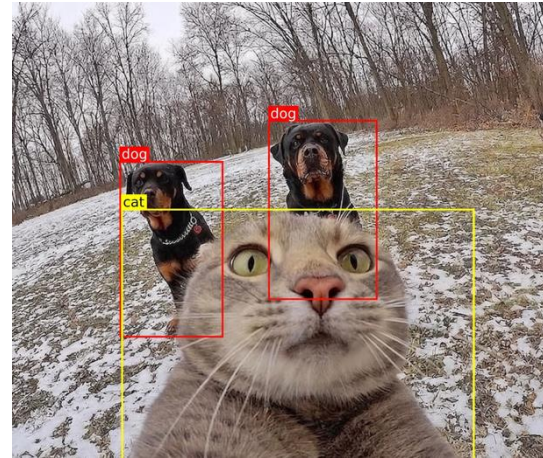
Semantic segmentation



Cat, Dog, Grass, Trees

Multiple objects,
No instance

Object detection



Cat, Dog

Multiple object,
Multiple instances

Instance segmentation



Cat, Dog, Dog

Image classification | reminder

Given an image and a set of classes {cat, dog, plane, chair, ...}:
Predict the class that best describes the image.



→ **Cat**

Computer vision tasks

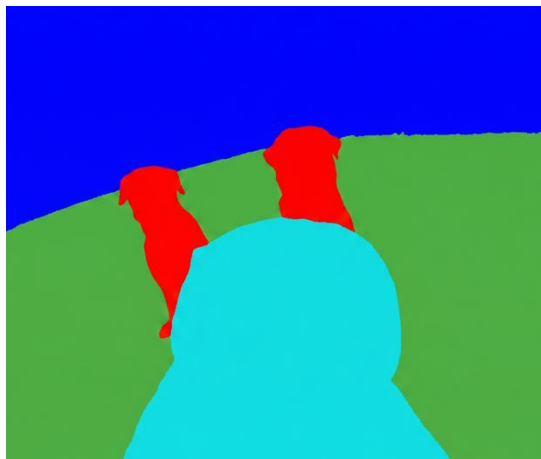
Image classification



Cat

No spatial context

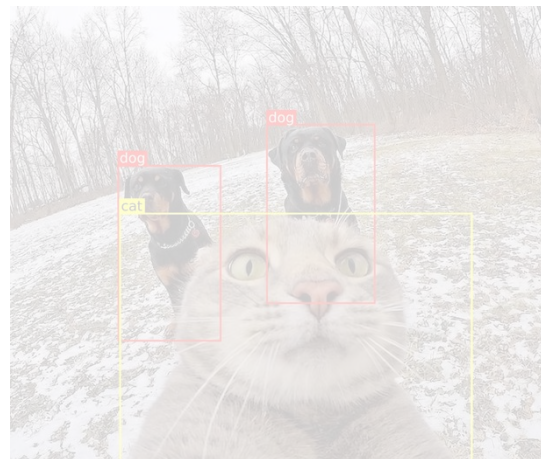
Semantic segmentation



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Semantic segmentation

Paired training data:

Given a set of classes {cat, dog, plane, chair, ...}, each pixel is labeled with a semantic category

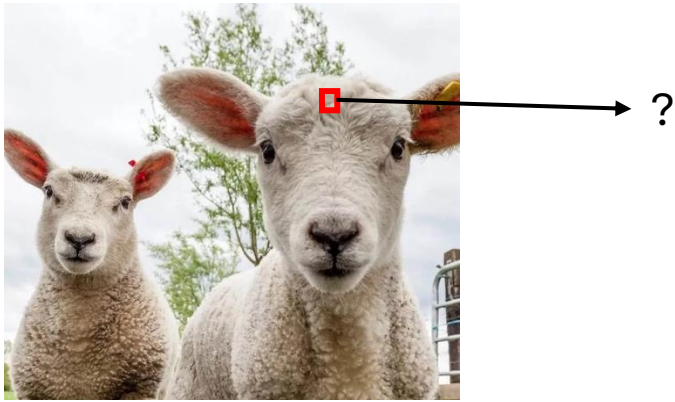


Cat, Dog, Grass, Trees



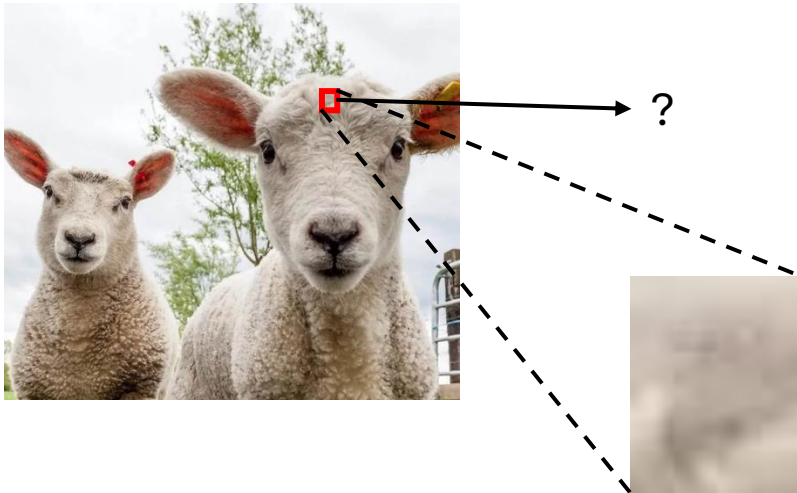
During inference: classify each pixel to a class

Semantic segmentation | sliding window

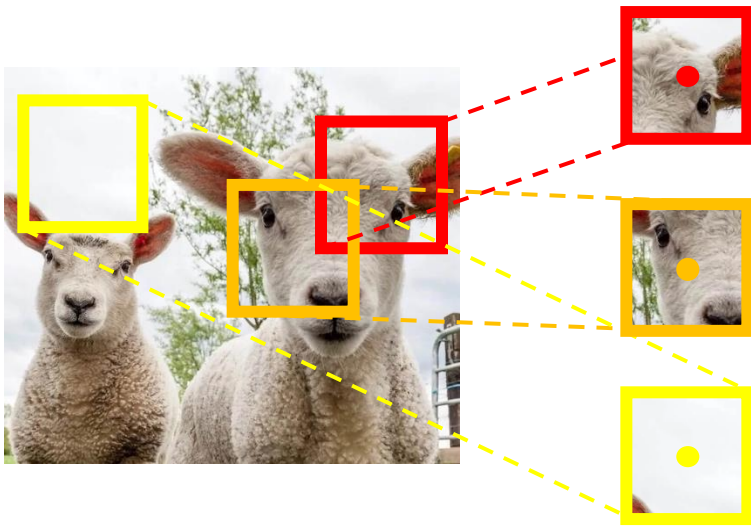


Semantic segmentation | sliding window

Impossible to do without context – so how do we include context?

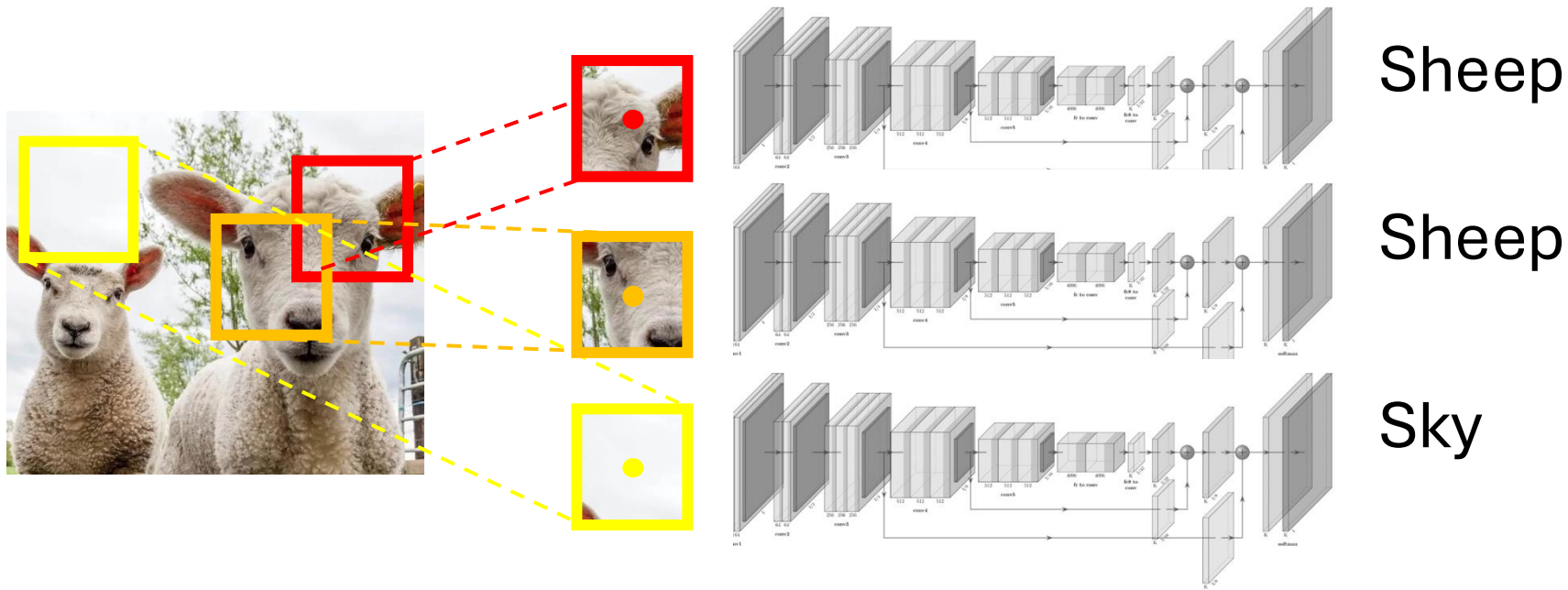


Semantic segmentation | sliding window



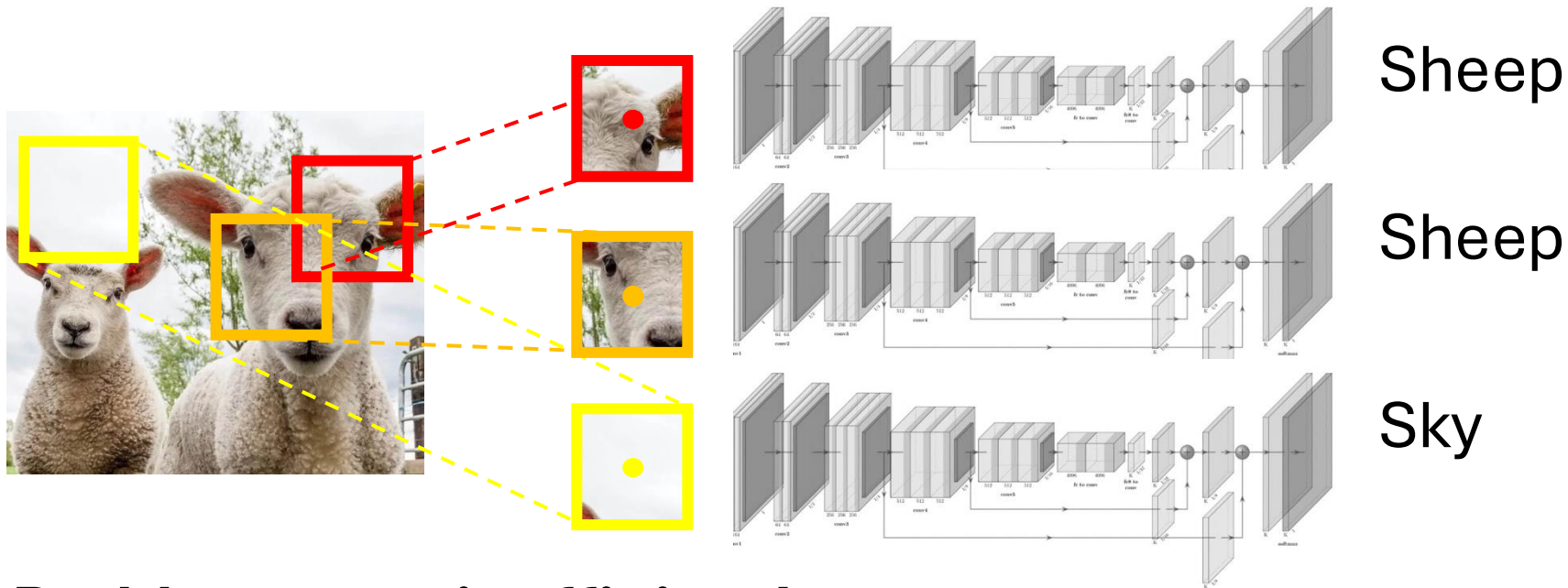
Semantic segmentation | sliding window

Classify center pixel with CNN



Semantic segmentation | sliding window

Classify center pixel with CNN

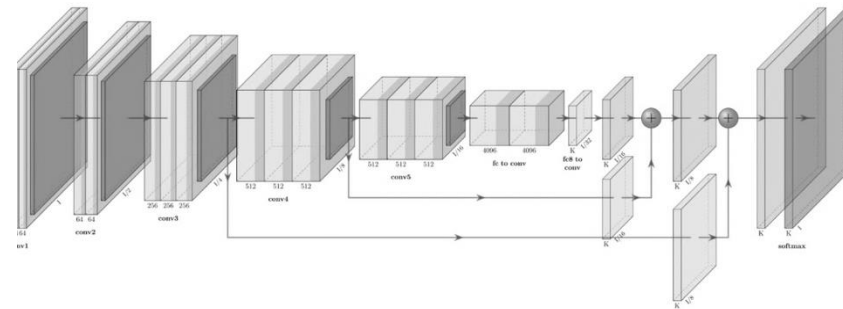


Problem: very inefficient!

Does not reuse shared features across overlapping patches.

Semantic segmentation | convolution

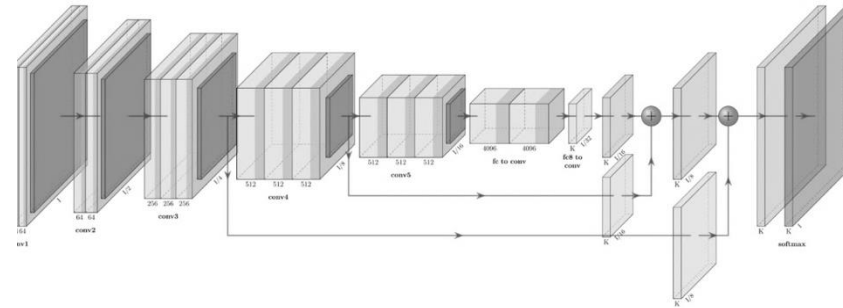
An idea: first encode with CNN, then perform semantic segmentation



Semantic segmentation | convolution

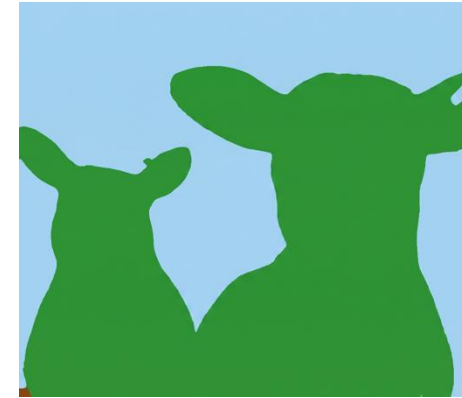
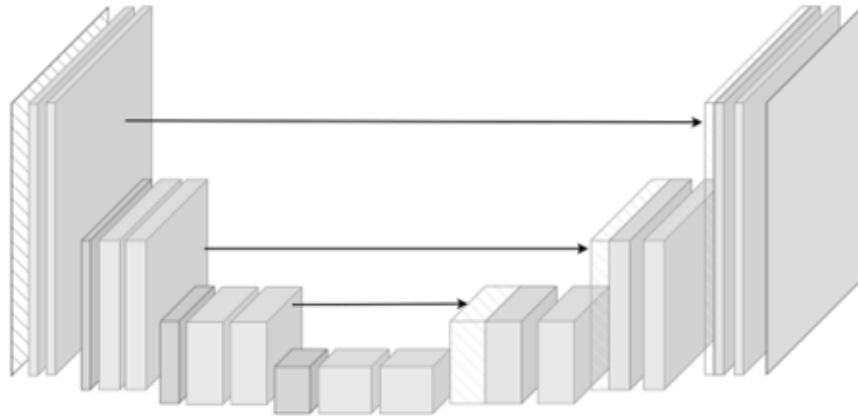
An idea: first encode with CNN, then perform semantic segmentation

The problem: the output should be the same size as the input but classification architectures usually reduce feature size.



Semantic segmentation | convolution

The solution: U-Net (Downsampling and upsampling)



Semantic segmentation



Each pixel is labeled with a category label,
No differentiation between instances.

Computer vision tasks

Image classification



Cat

No spatial context

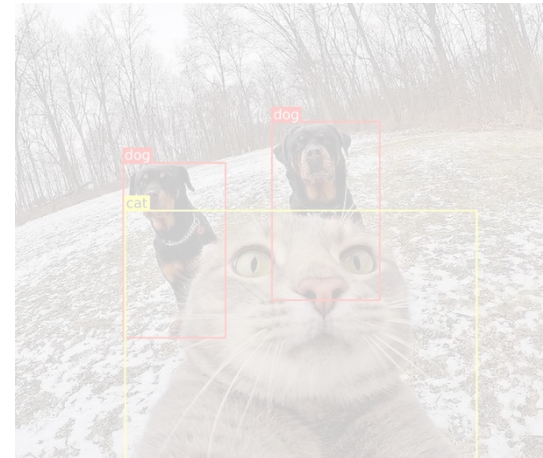
Semantic segmentation



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No instance

Object detection



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Cat, Dog, Dog

Computer vision tasks

Image classification



Cat

No spatial context

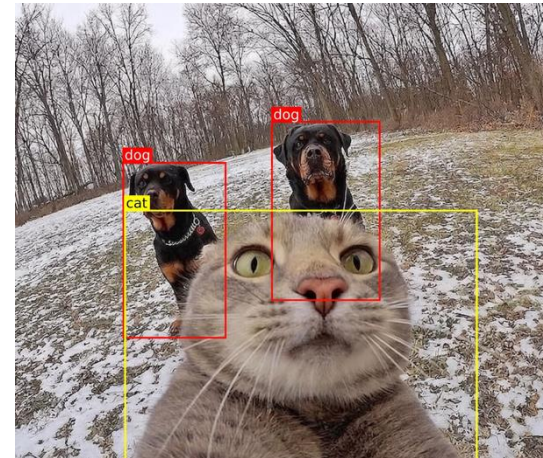
Semantic segmentation



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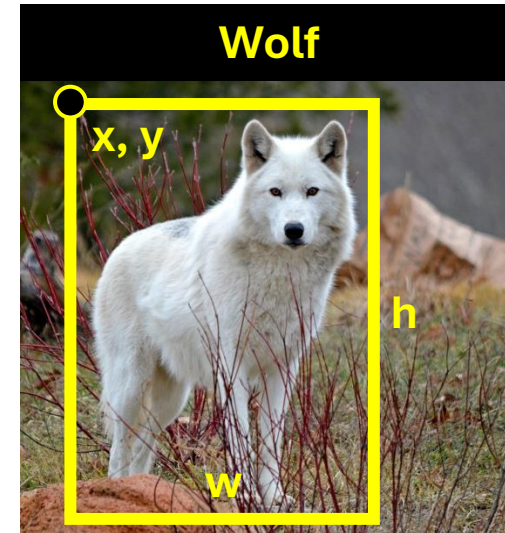
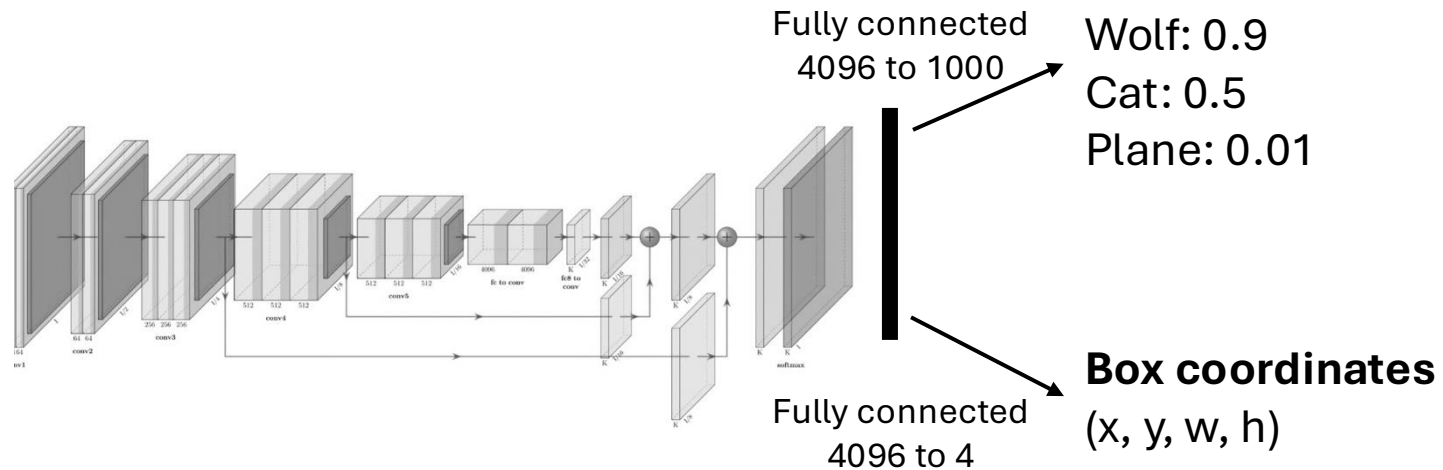
Instance segmentation



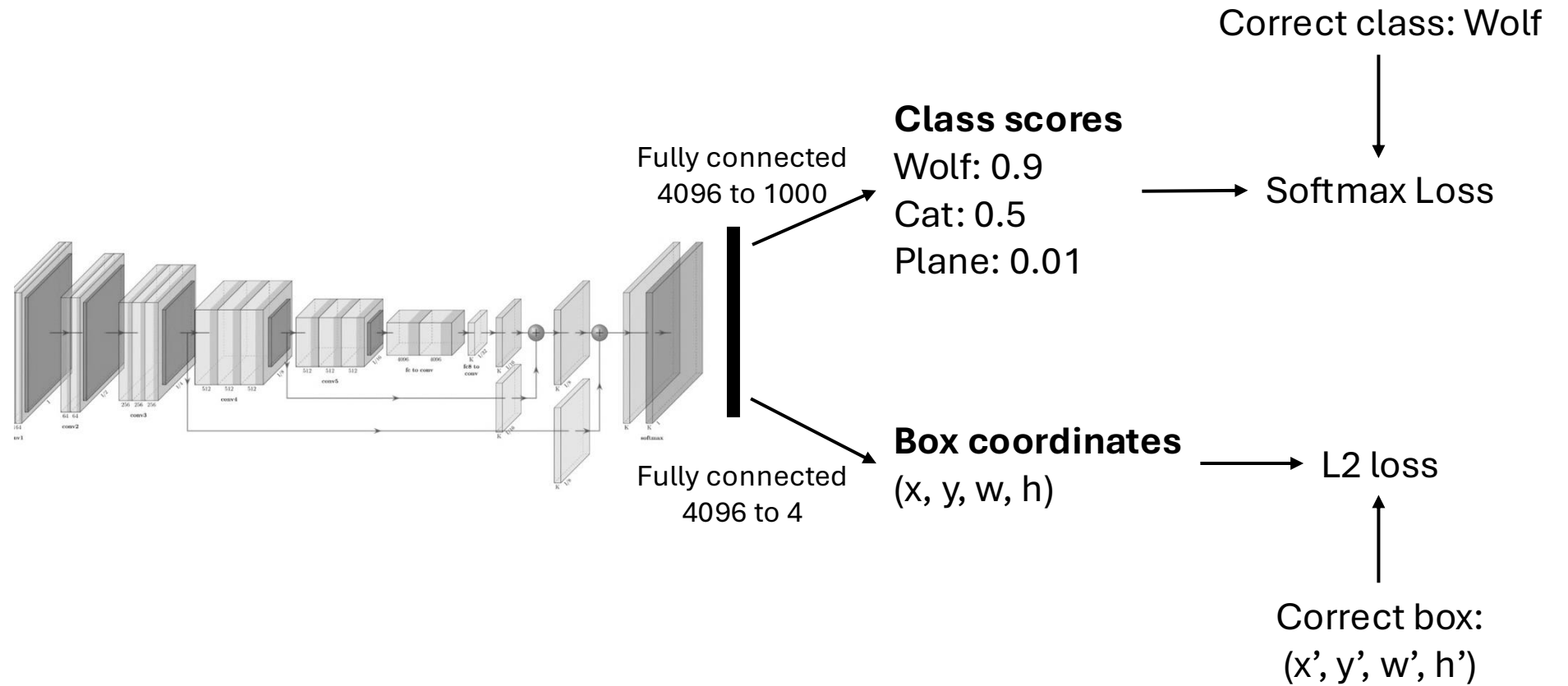
Cat, Dog, Dog

Object detection | single object

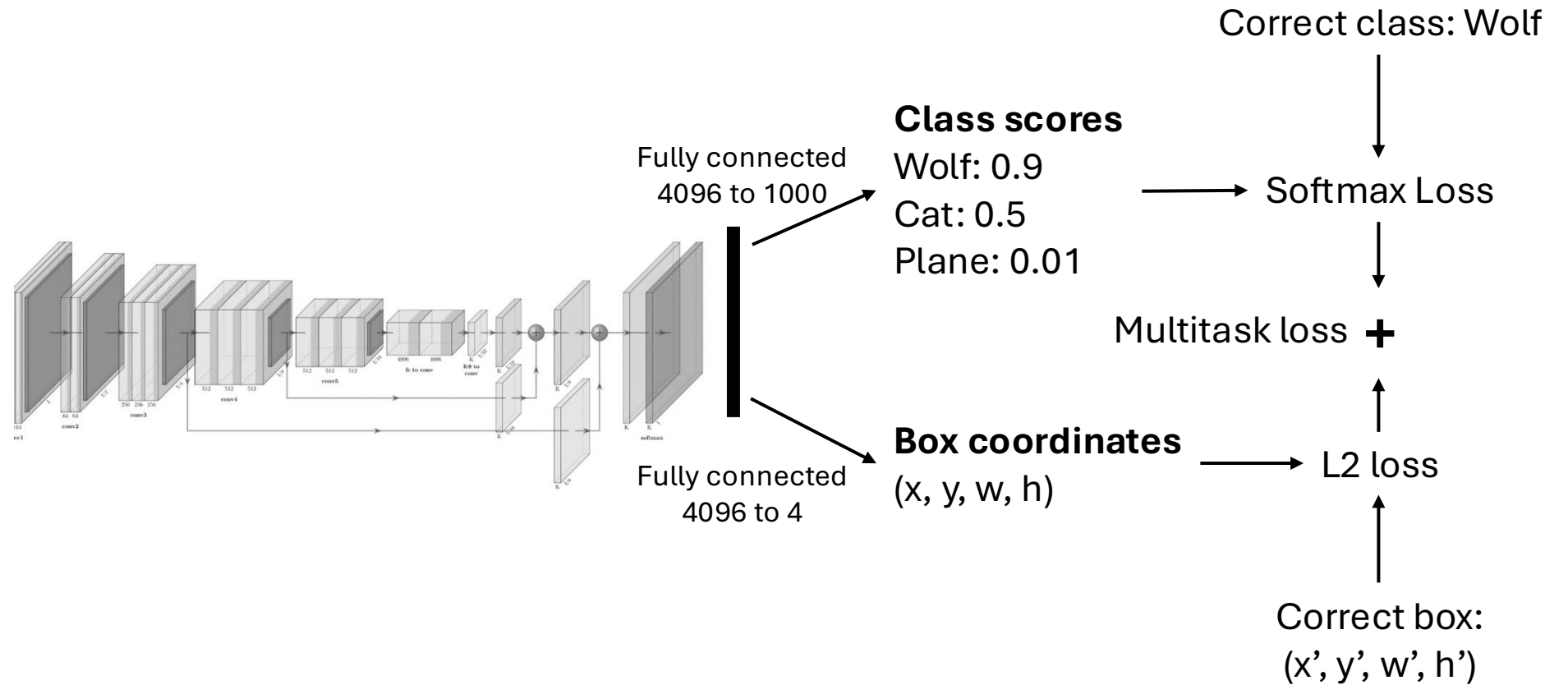
Performs classification + localization



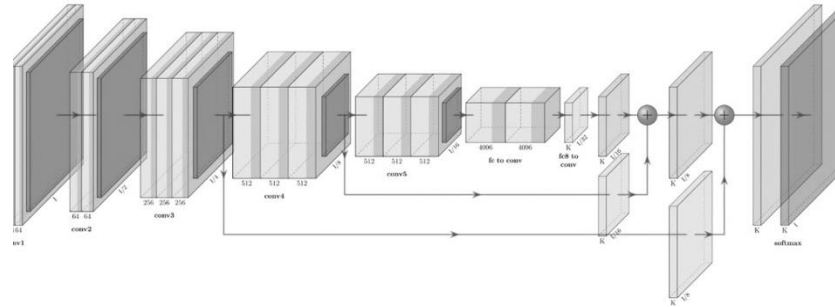
Object detection | single object



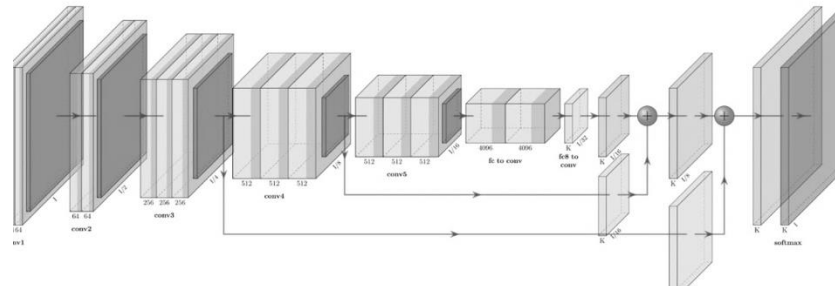
Object detection | single object



Object detection | multiple objects

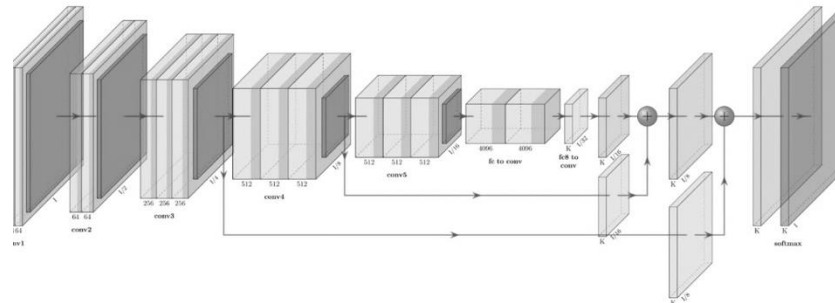


Wolf, (x, y, w, h)



Sheep, (x, y, w, h)

Sheep, (x, y, w, h)



Zebra, (x, y, w, h)

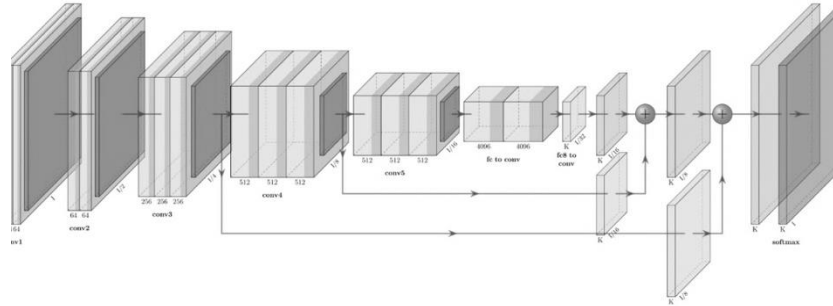
Zebra, (x, y, w, h)

Zebra, (x, y, w, h)

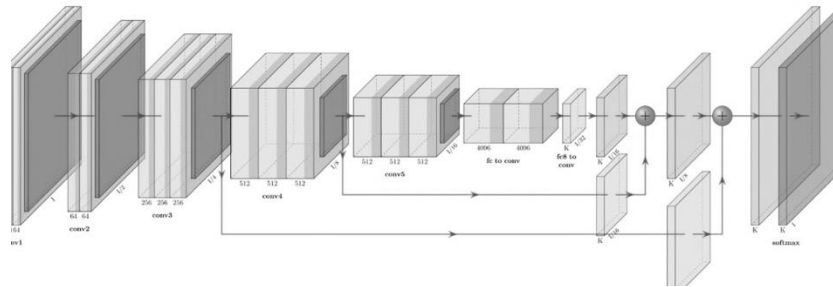
...

Object detection | multiple objects

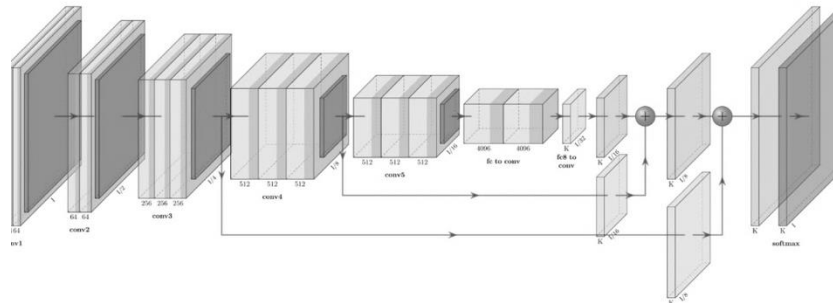
Problem: each image has a different number of output!



Wolf, (x, y, w, h)
[1 class and 4 numbers]



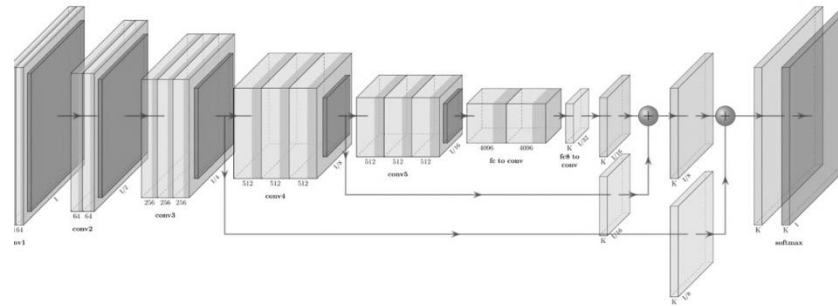
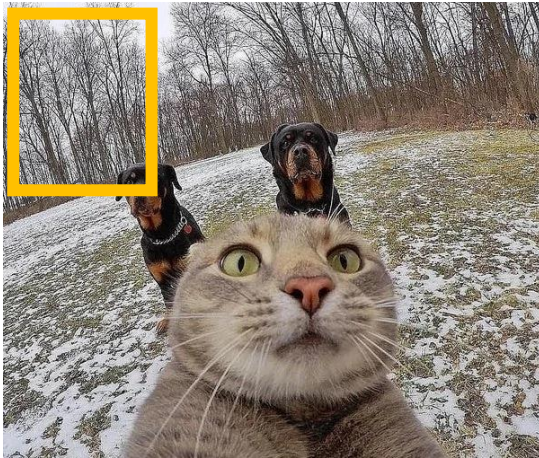
Sheep, (x, y, w, h)
Sheep, (x, y, w, h)
[2 classes and 8 numbers]



Zebra, (x, y, w, h)
Zebra, (x, y, w, h)
Zebra, (x, y, w, h)
...
[Many classes and many numbers]

Object detection | multiple objects

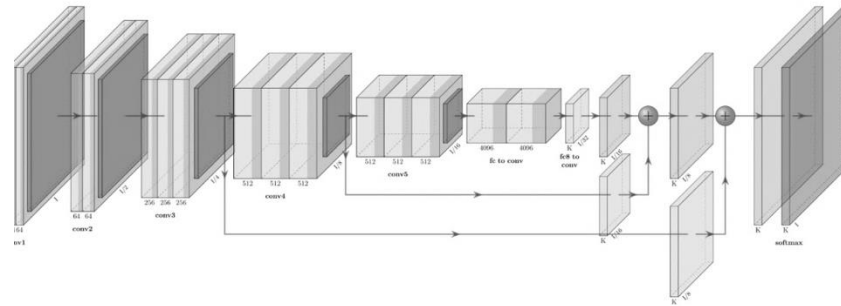
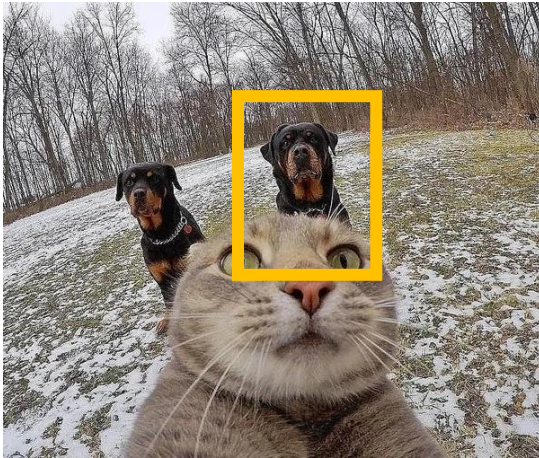
Apply a CNN to many patches and classify as class or background



Dog?	No
Cat?	No
Background?	Yes

Object detection | multiple objects

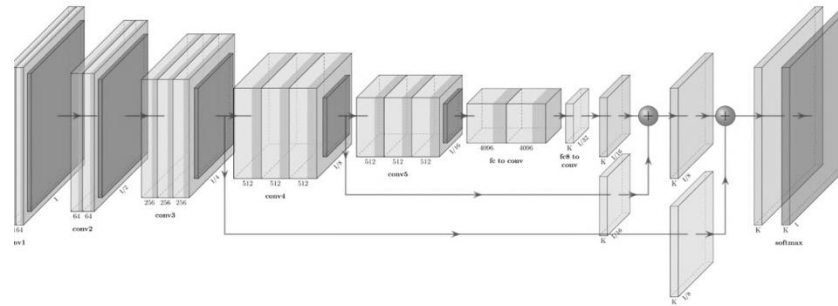
Apply a CNN to many patches and classify as class or background



Dog?	Yes
Cat?	No
Background?	No

Object detection | multiple objects

Apply a CNN to many patches and classify as class or background

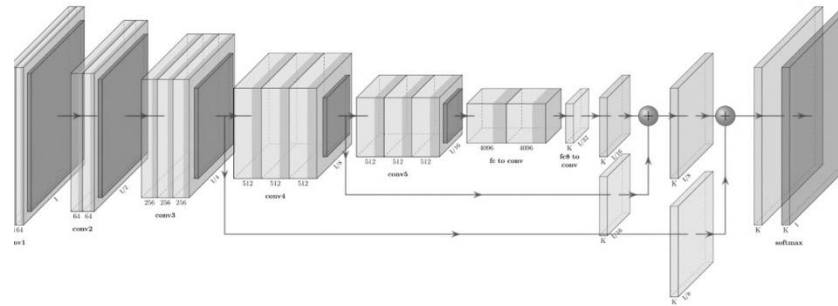


Dog?	No
Cat?	Yes
Background?	No

Object detection | multiple objects

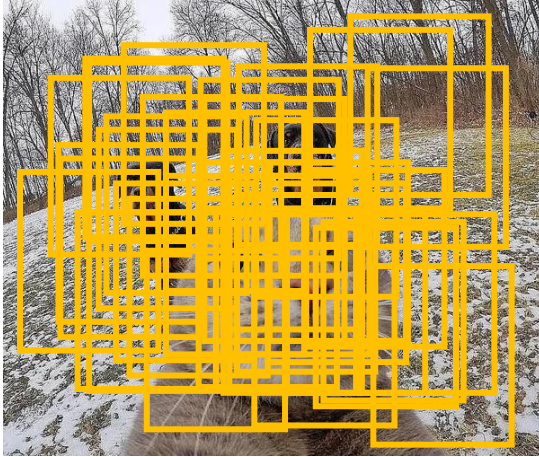
Apply a CNN to many patches and classify as class or background

What is the problem?



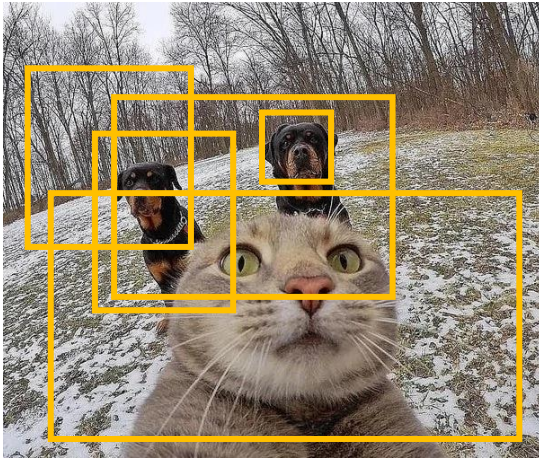
Dog?	No
Cat?	Yes
Background?	No

Object detection | multiple objects



Need to apply a CNN to a huge number of locations, scales and aspect ratios
Very computationally expensive!

Region proposal | selective search



Find rough image regions that are likely to contain objects. Relatively fast to run, selective search yields ~2000 region proposals in a few second on CPU.

Measuring the Objectness of Image Windows. Alexe et al. 2012

Selective Search for Object Recognition. Uijlings et al. 2013

BING: Binarized Normed Gradients for Objectness Estimation at 300 FPS. Cheng et al. 2014

Edge Boxes: Locating Object Proposals From Edges. Zitnick and Dollar. 2014

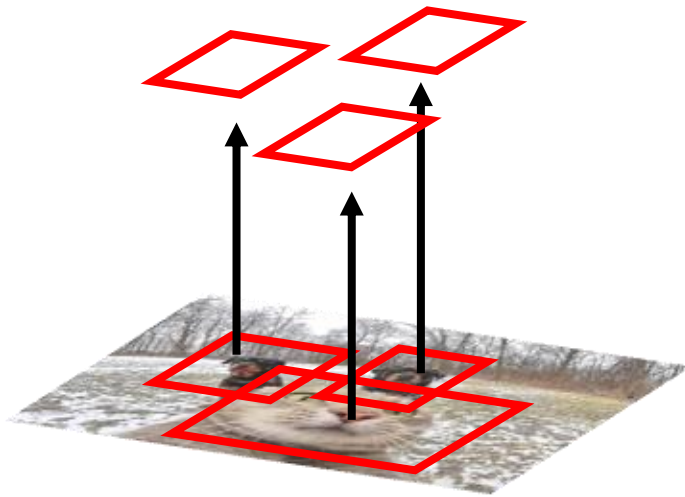
MCV 97202 | Technion | Spring 26

R(egion)-CNN



Regions of Interest (RoI)
from a proposal method

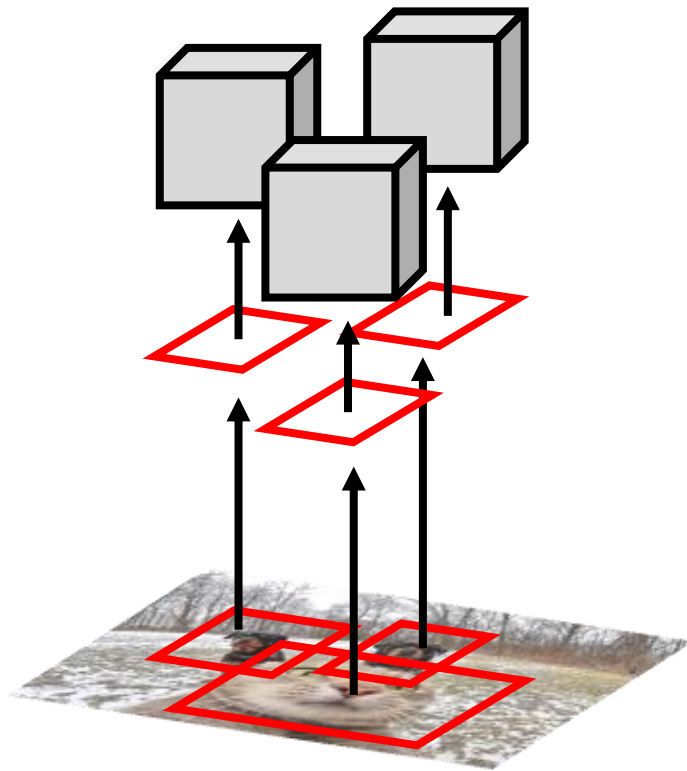
R-CNN



Warped image regions
(224 x 224 pixels)

Regions of Interest (RoI)
from a proposal method

R-CNN

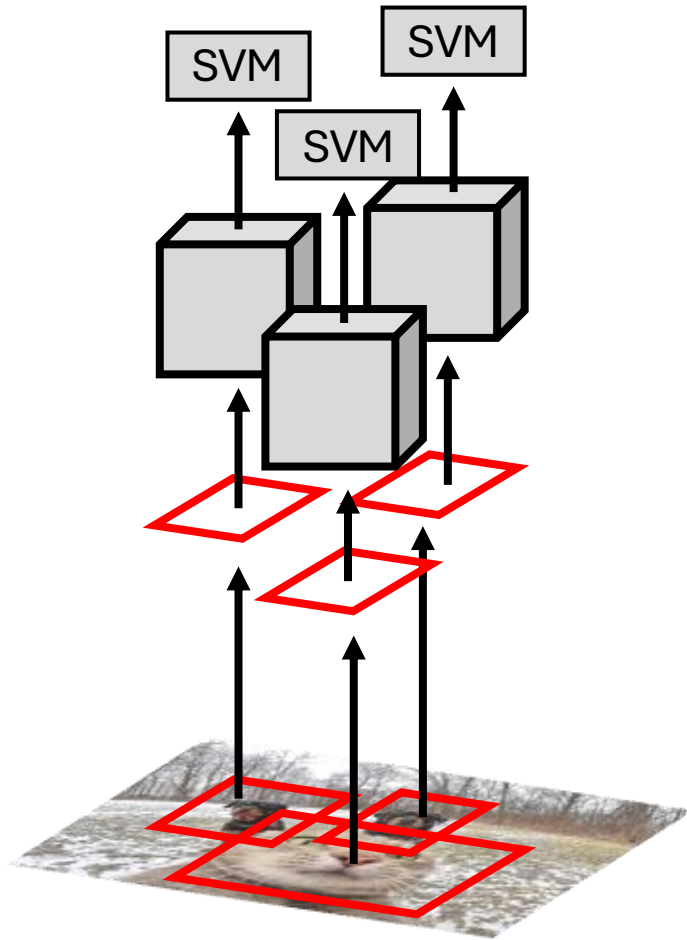


Forward each region through a CNN

Warped image regions
(224 x 224 pixels)

Regions of Interest (RoI)
from a proposal method

R-CNN



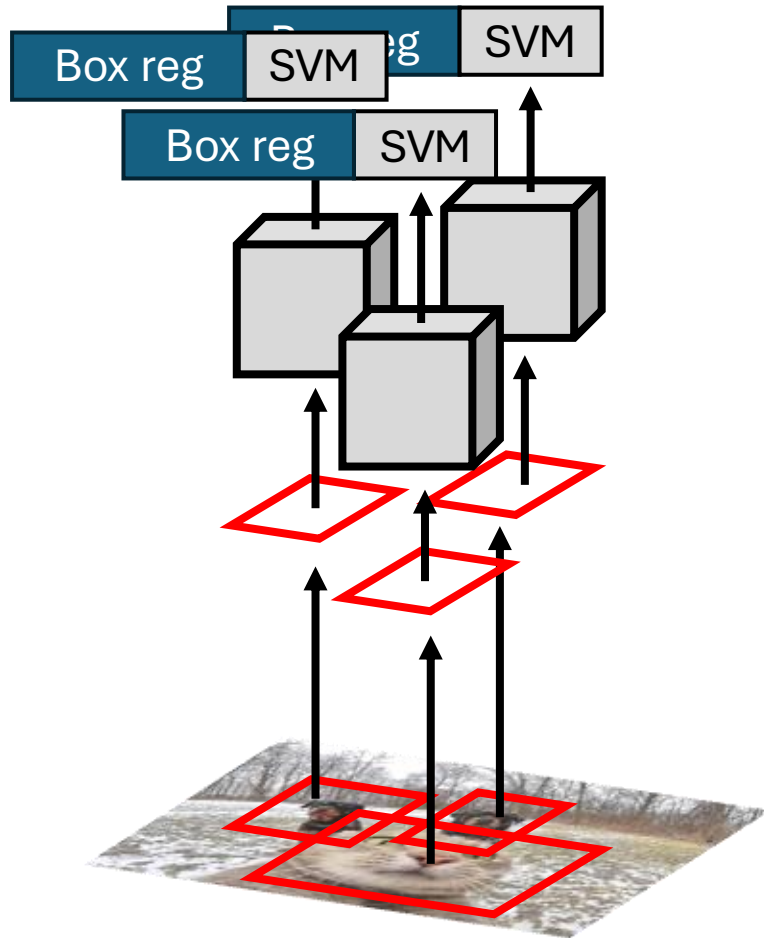
Classify each region with SVM

Forward each region through a CNN

Warped image regions
(224 x 224 pixels)

Regions of Interest (RoI)
from a proposal method

R-CNN



Classify each region with SVM

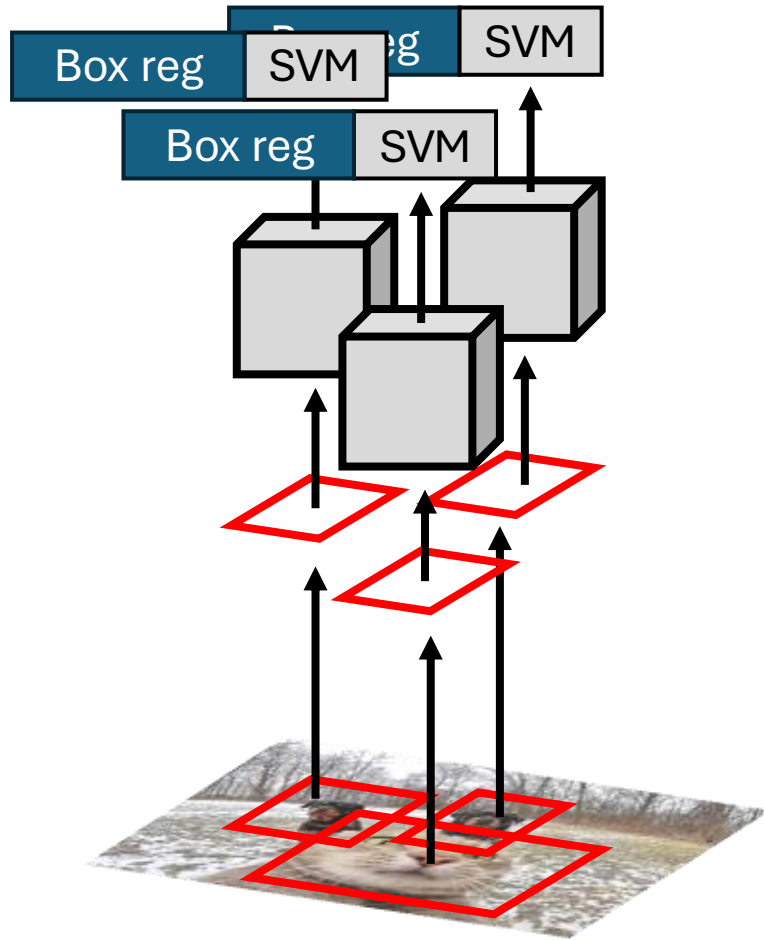
Regress correction to the RoI
(dx, dy, dw, dh)

Forward each region through a CNN

Warped image regions
(224 x 224 pixels)

Regions of Interest (RoI)
from a proposal method

R-CNN



Classify each region with SVM

Regress correction to the RoI
(dx, dy, dw, dh)

Forward each region through a CNN

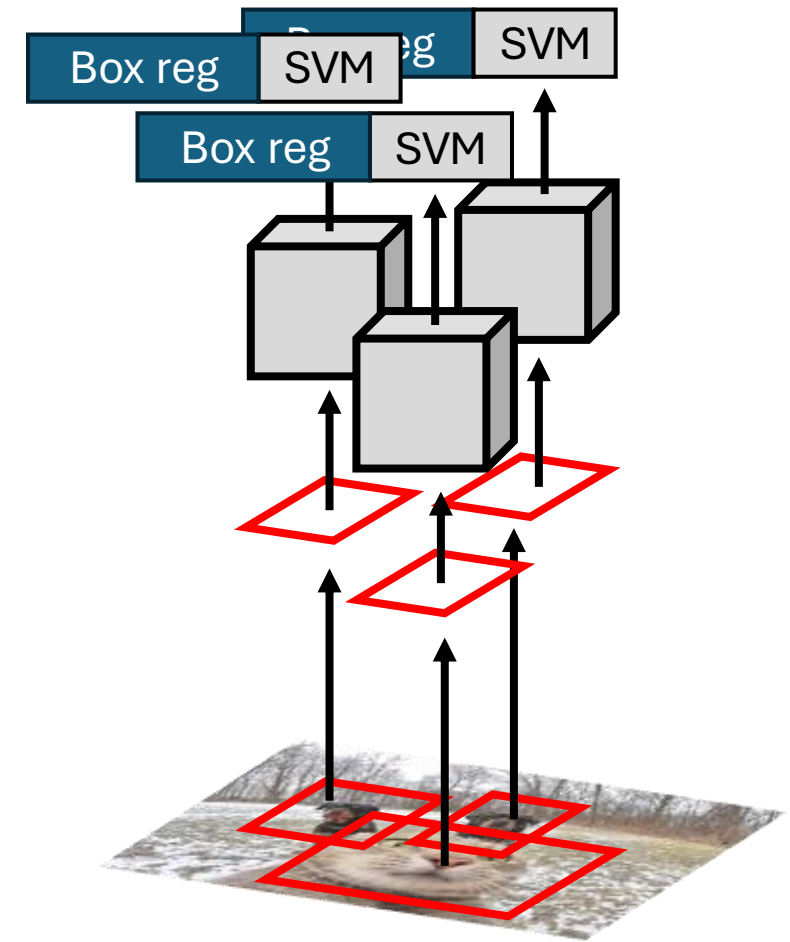
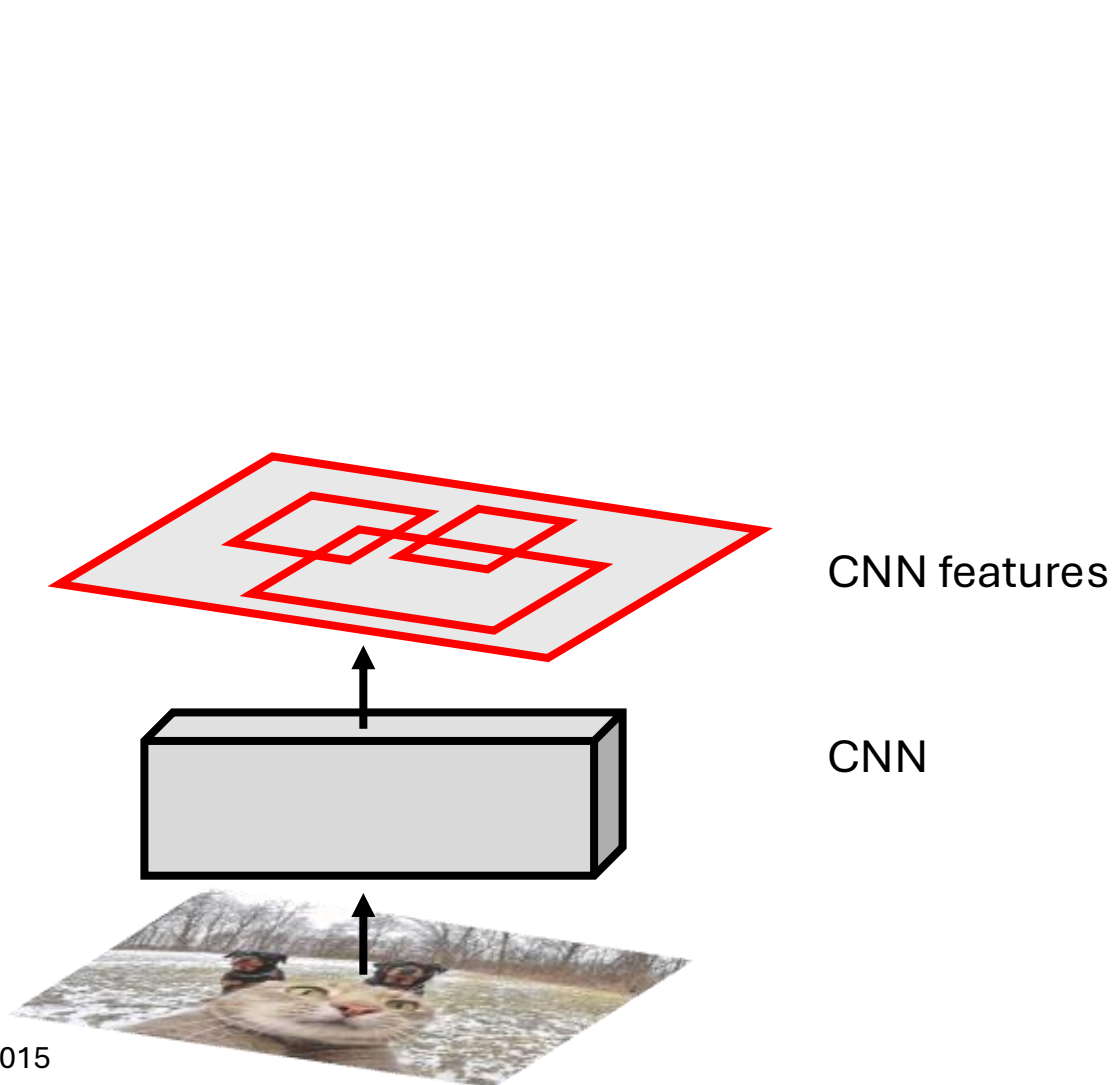
Warped image regions
(224 x 224 pixels)

Regions of Interest (RoI)
from a proposal method

**Problem: need to do ~2000
independent forward passes per
image.
Very slow!**

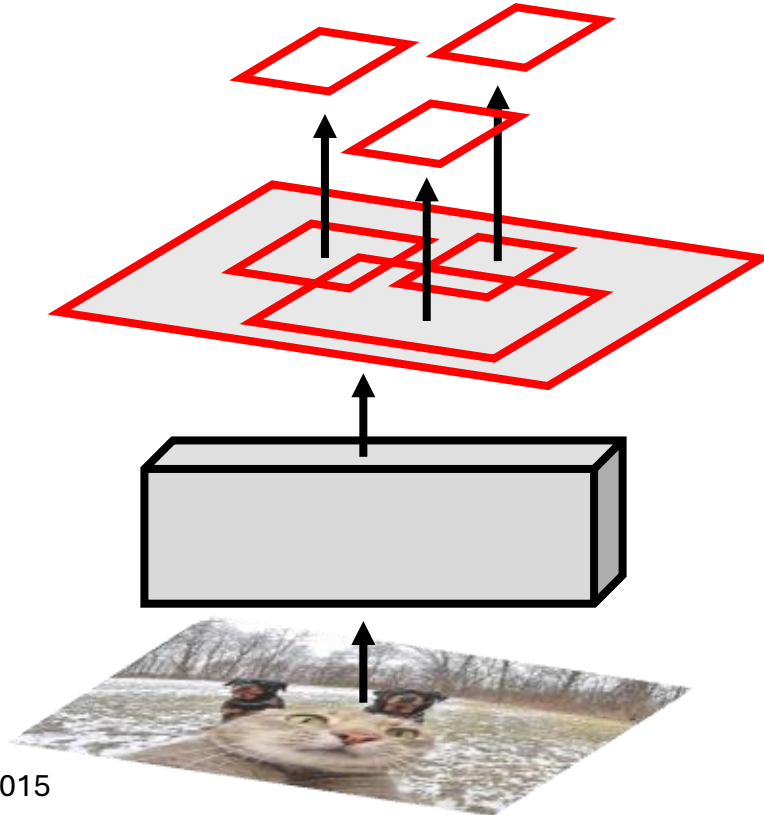
Solution: pass the image through CNN before cropping.

Fast R-CNN



Solution: pass the image through CNN before cropping.

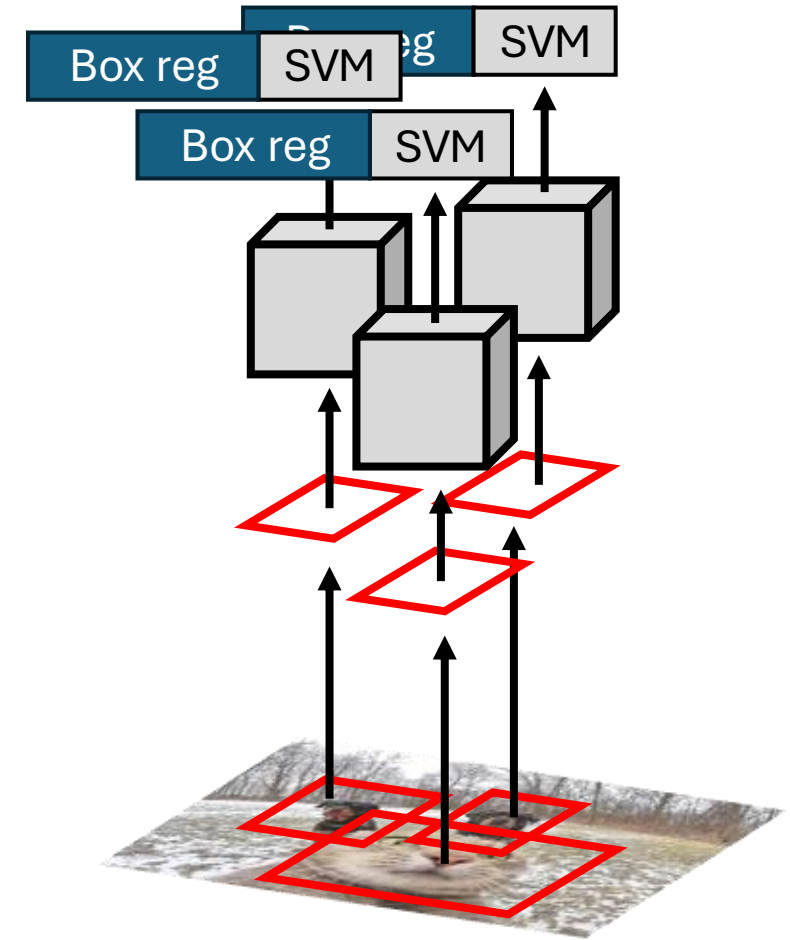
Fast R-CNN



Warped regions

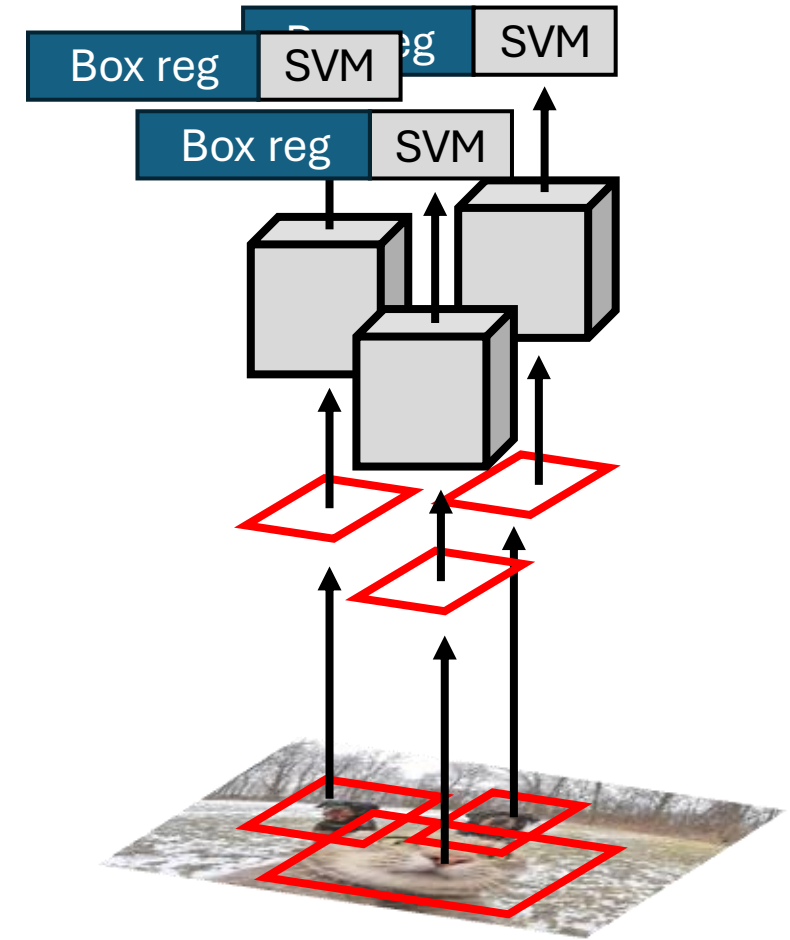
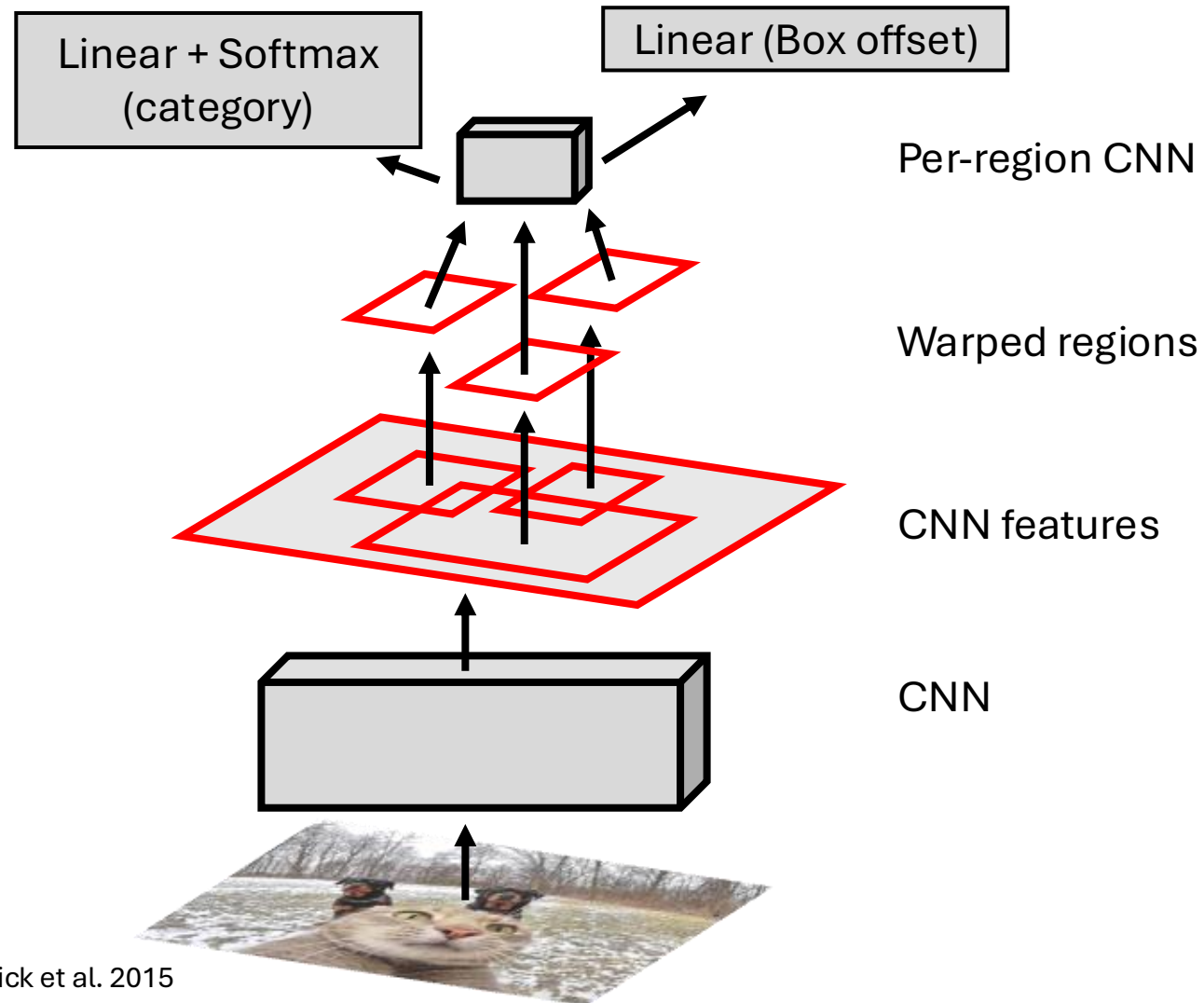
CNN features

CNN



Solution: pass the image through CNN before cropping.

Fast R-CNN

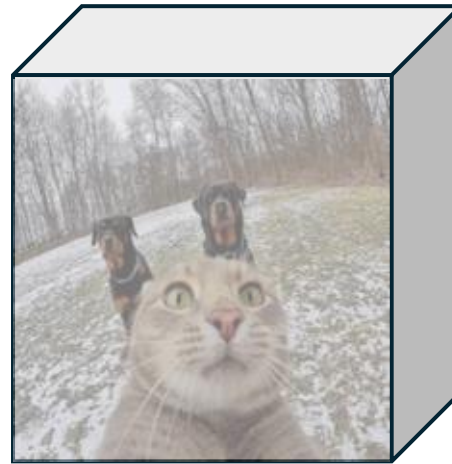


Region proposal network



3 x 640 x 480

CNN



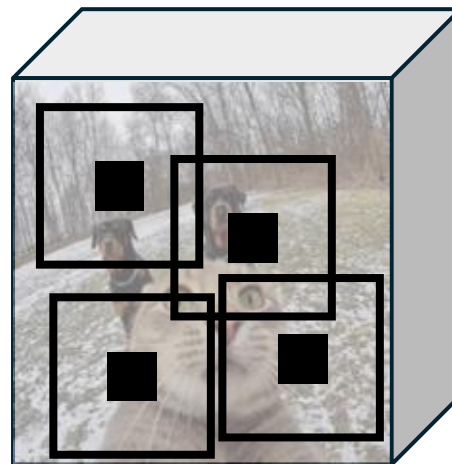
512 x 20 x 15

Region proposal network



3 x 640 x 480

CNN



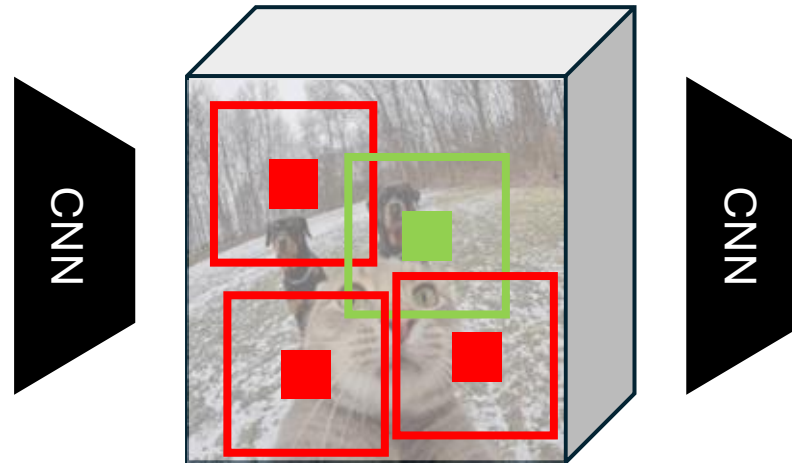
512 x 20 x 15

An anchor box of fixed size at each point in the feature map

Region proposal network



3 x 640 x 480

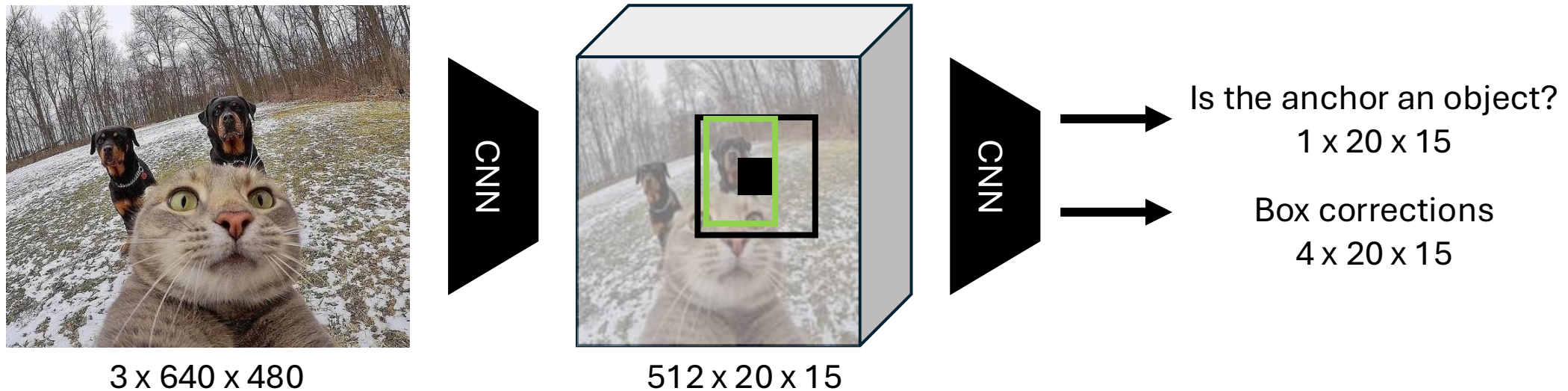


512 x 20 x 15

Is the anchor an object?
1 x 20 x 15

**For each point, predict whether
the anchor contains an object**

Region proposal network

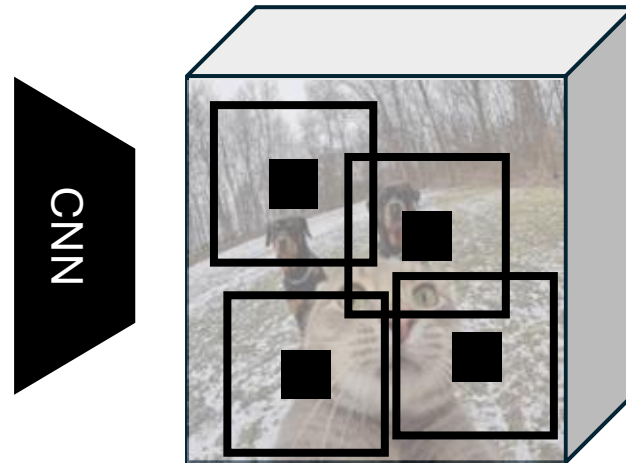


For positive boxes, predict the correction from the anchor box

Region proposal network



3 x 640 x 480



512 x 20 x 15



Is the anchor an object?

$K \times 20 \times 15$

Box corrections

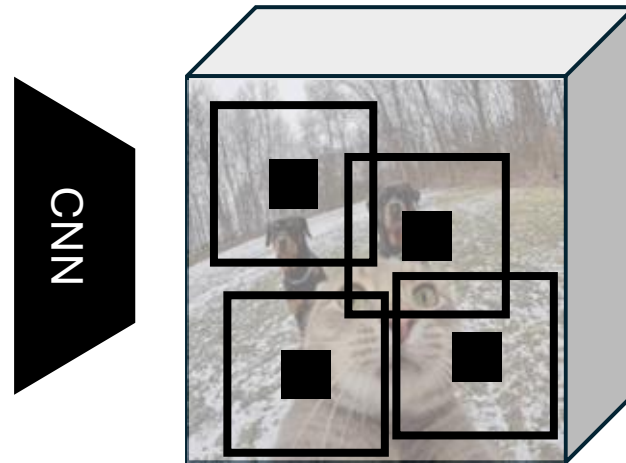
$4K \times 20 \times 15$

**In practice uses K anchor
boxes of different size / scale**

Region proposal network



3 x 640 x 480



512 x 20 x 15



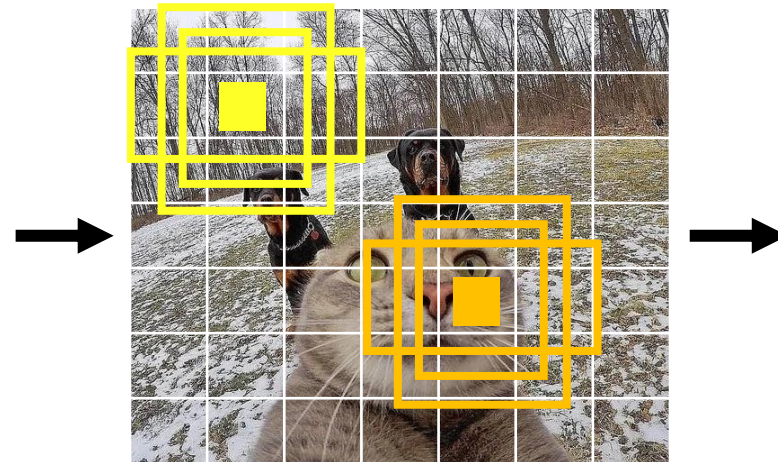
Is the anchor an object?
 $K \times 20 \times 15$

Box corrections
 $4K \times 20 \times 15$

In practice uses K anchor boxes of different size / scale
The $K \times 20 \times 15$ boxes are sorted by “objectness” and the top ~300 are taken

Single-stage object detectors

YOLO / SSD / RetinaNet



Divide into a grid
 7×7

Set B boxes centered at each grid cell
Here $B = 3$

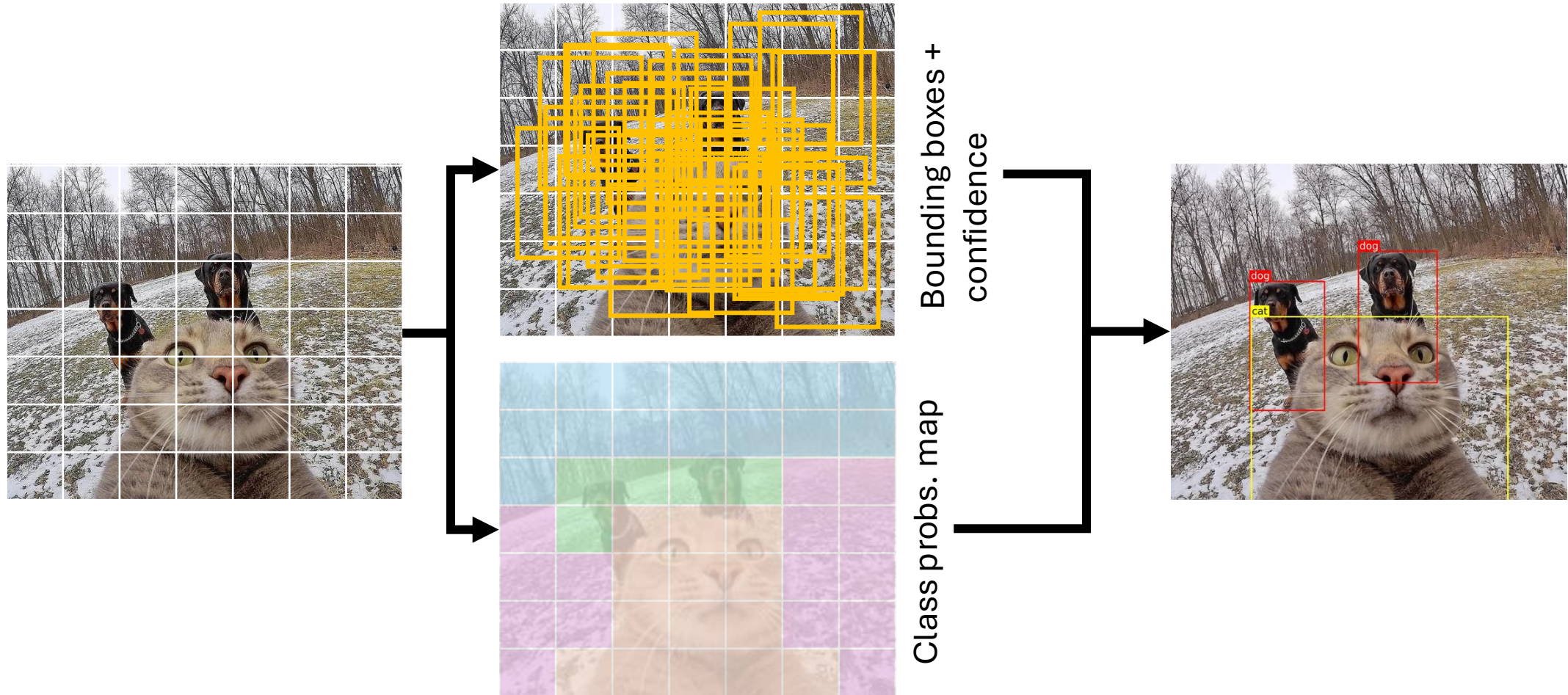
Within each grid cell:

- Regress from each of the B boxes to a final box with 5 numbers: (dx, dy, dw, dh, confidence).
- Predict scores for each of the C classes (incl. background).

Like PRN, but category specific.

Output shape:
 $7 \times 7 \times (5B + C)$

YOLO | You Only Look Once



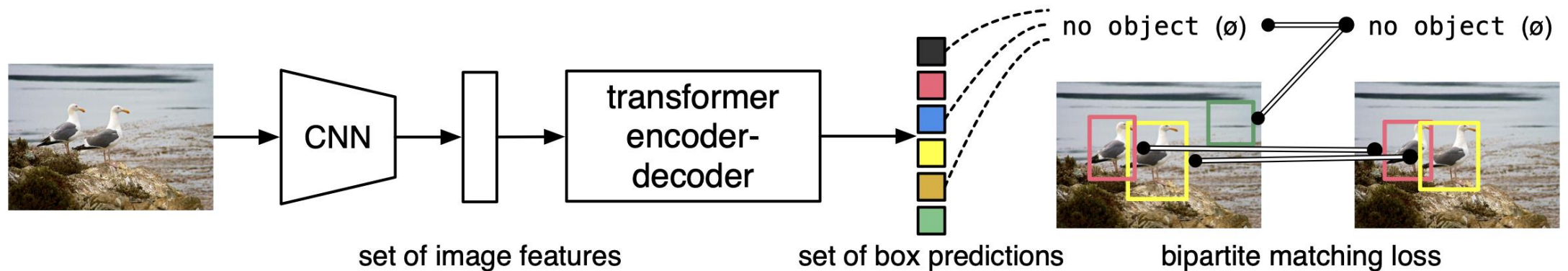
Object **D**etection with **T**Ransformers

DETR

Directly output a set of boxes from a Transformer.

No anchors, no regression of box transforms to rough box coordinates.

Matches object queries to encoded image patches, trained to regress box coordinates through a bipartite matching loss.

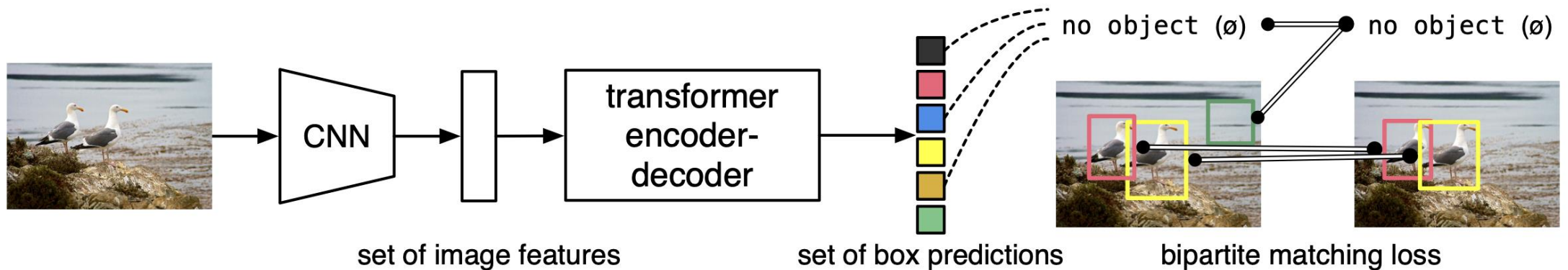


Object **D**etection with **T**Ransformers

DETR

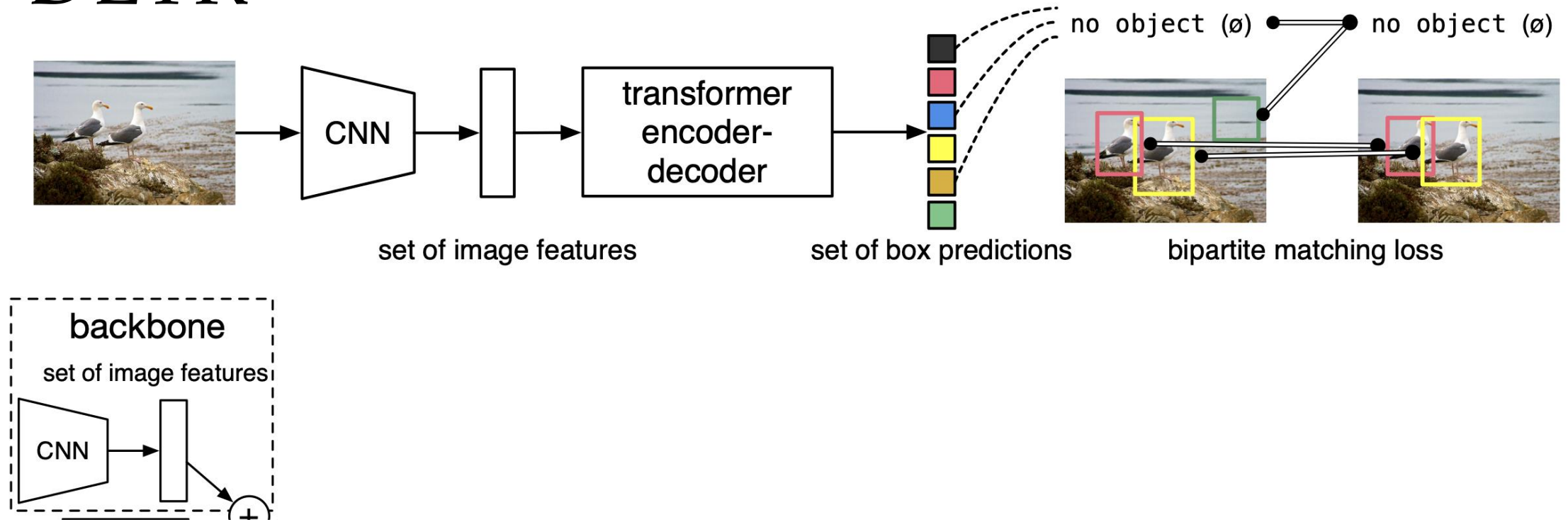
$$\mathcal{L}_{\text{Hungarian}}(y, \hat{y}) = \sum_{i=1}^N \left[-\log \hat{p}_{\hat{\sigma}(i)}(c_i) + \mathbb{I}_{\{c_i \neq \emptyset\}} \mathcal{L}_{\text{box}}(b_i, \hat{b}_{\hat{\sigma}(i)}) \right]$$

$$\mathcal{L}_{\text{box}}(b_i, \hat{b}_{\hat{\sigma}(i)}) = \lambda_{\text{IoU}} \mathcal{L}_{\text{IoU}}(b_i, \hat{b}_{\hat{\sigma}(i)}) + \lambda_{L1} \|b_i - \hat{b}_{\hat{\sigma}(i)}\|_1$$



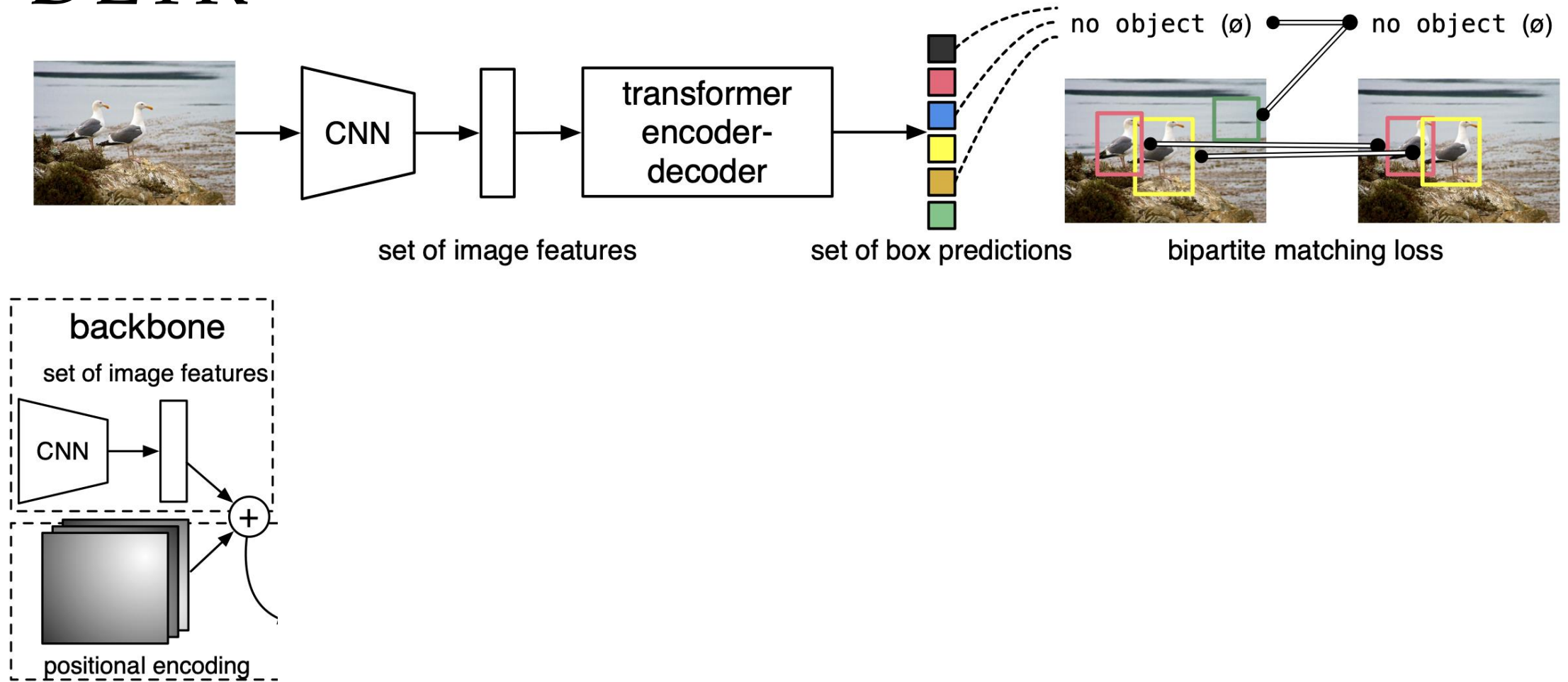
Object **D**etection with **T**Ransformers

DETR



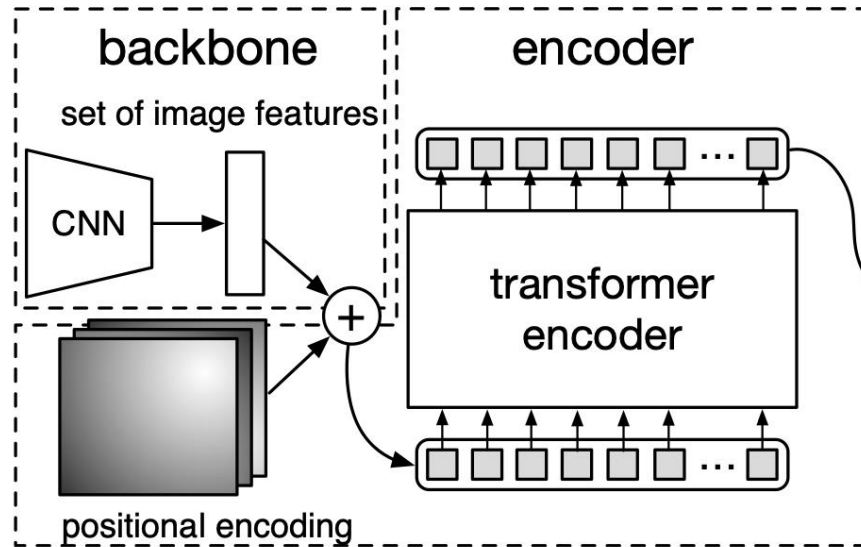
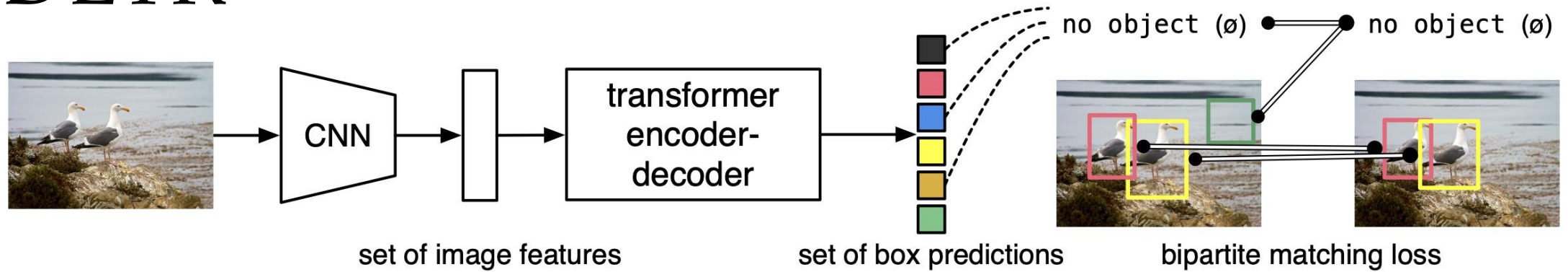
Object D_Etection with T_Ransformers

DETR



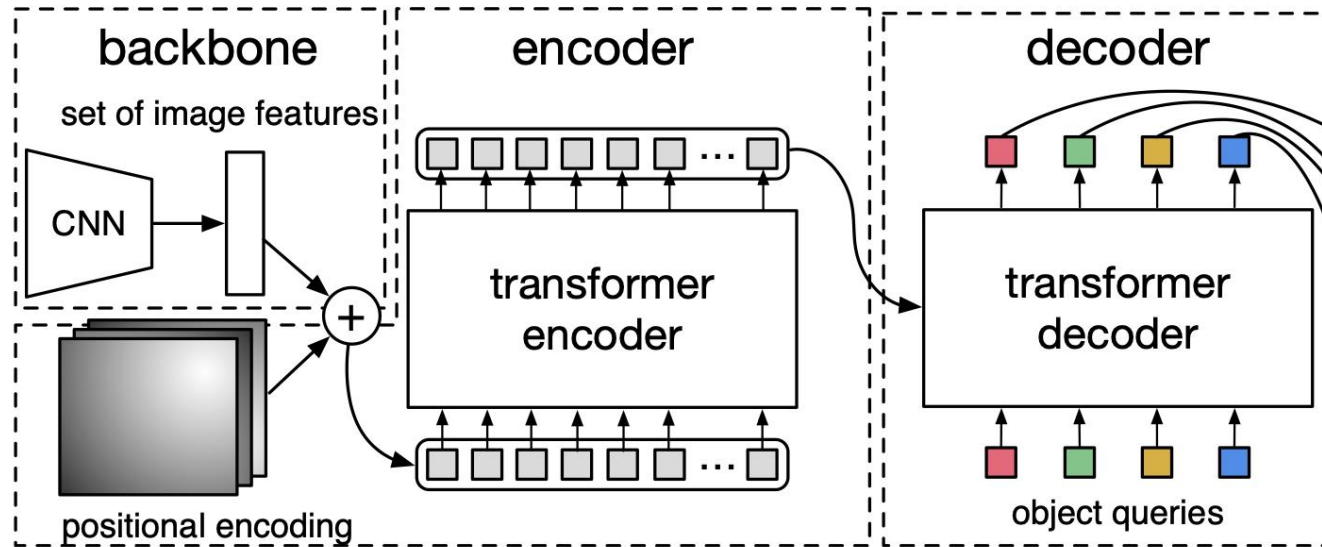
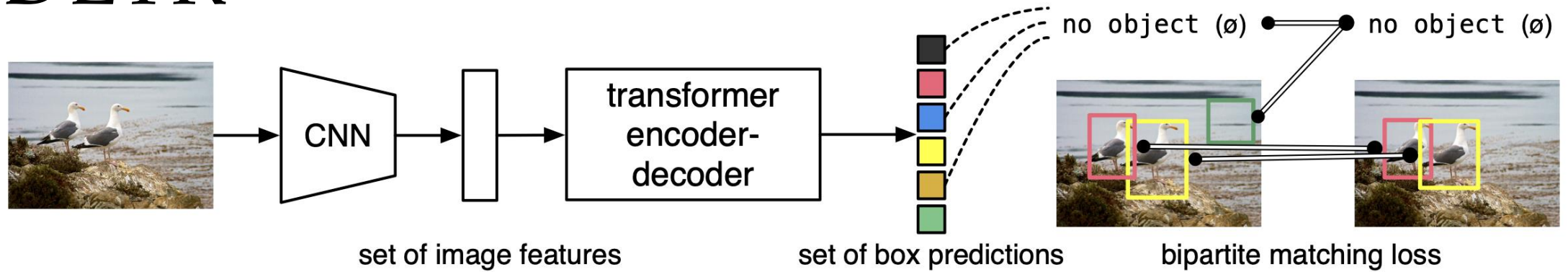
Object D_Etection with T_Ransformers

DETR



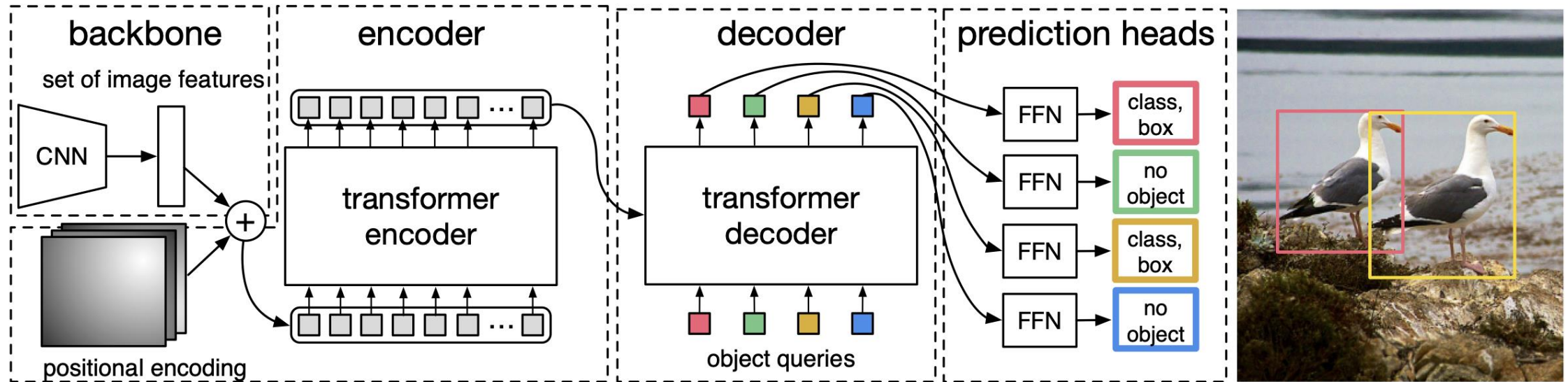
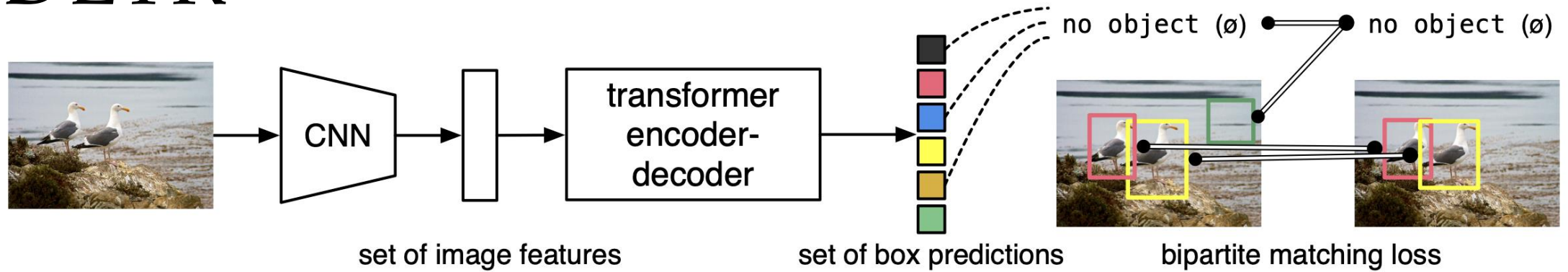
Object **D**etection with **T**Ransformers

DETR



Object **D**etection with **T**Ransformers

DETR



Computer vision tasks

Image classification



Cat

No spatial context

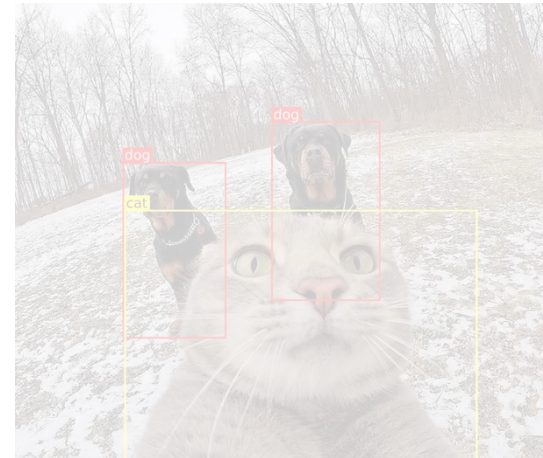
Semantic segmentation



Cat, Dog, Grass, Trees

Multiple objects,
No instance

Object detection



Cat, Dog

Multiple object,
Multiple instances

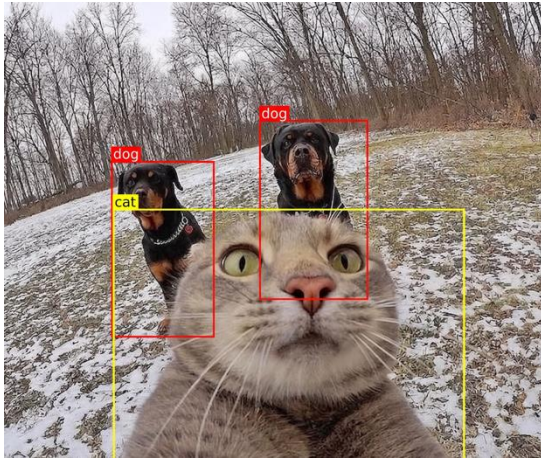
Instance segmentation



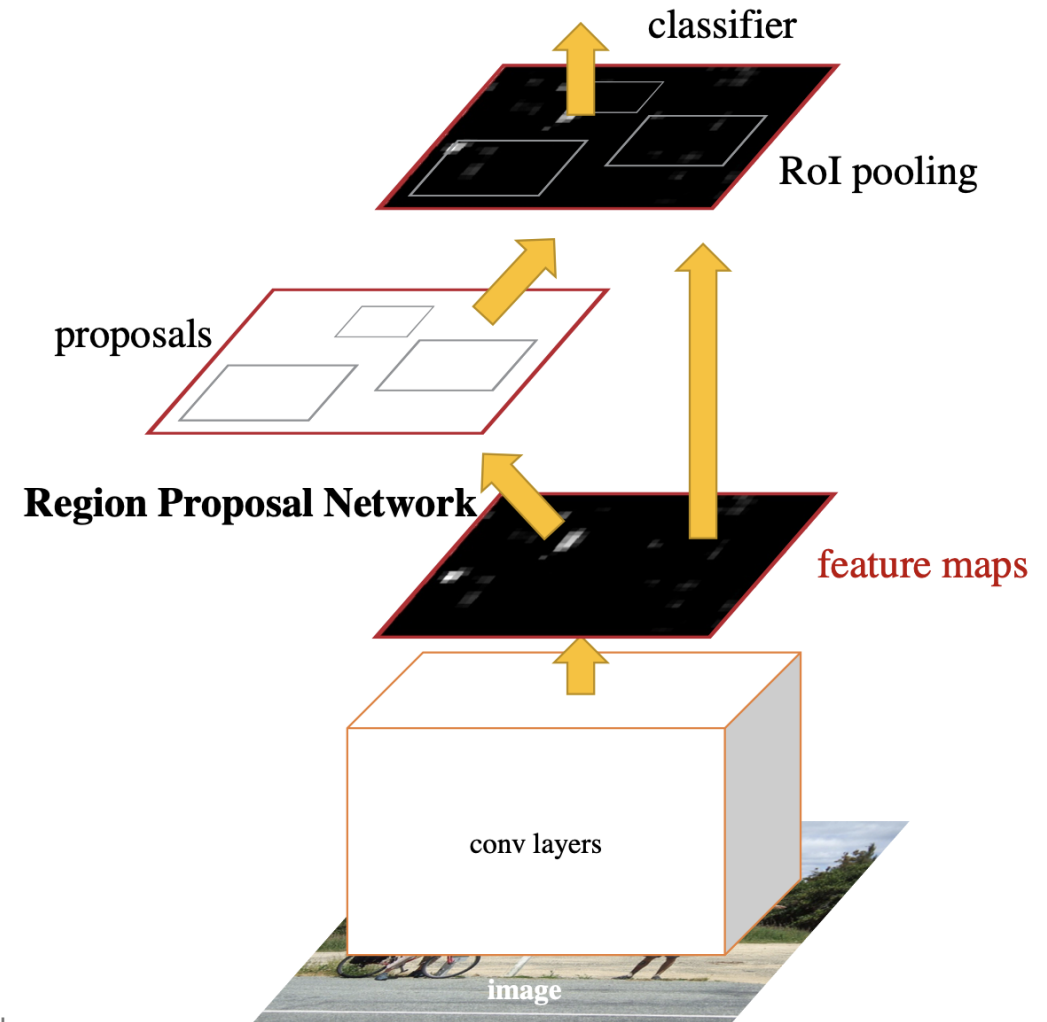
Cat, Dog, Dog

Object detection | Faster R-CNN

Object detection

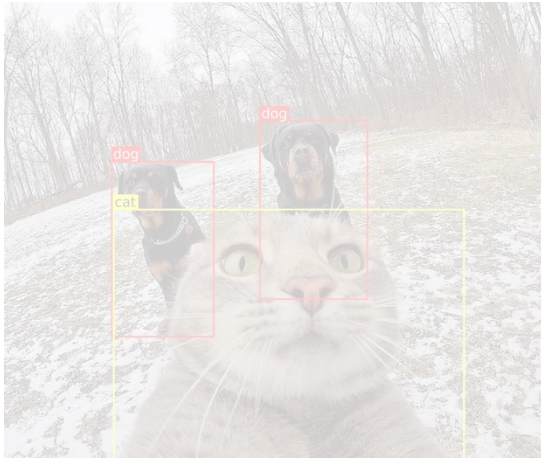


Instance segmentation



Instance segmentation | Mask R-CNN

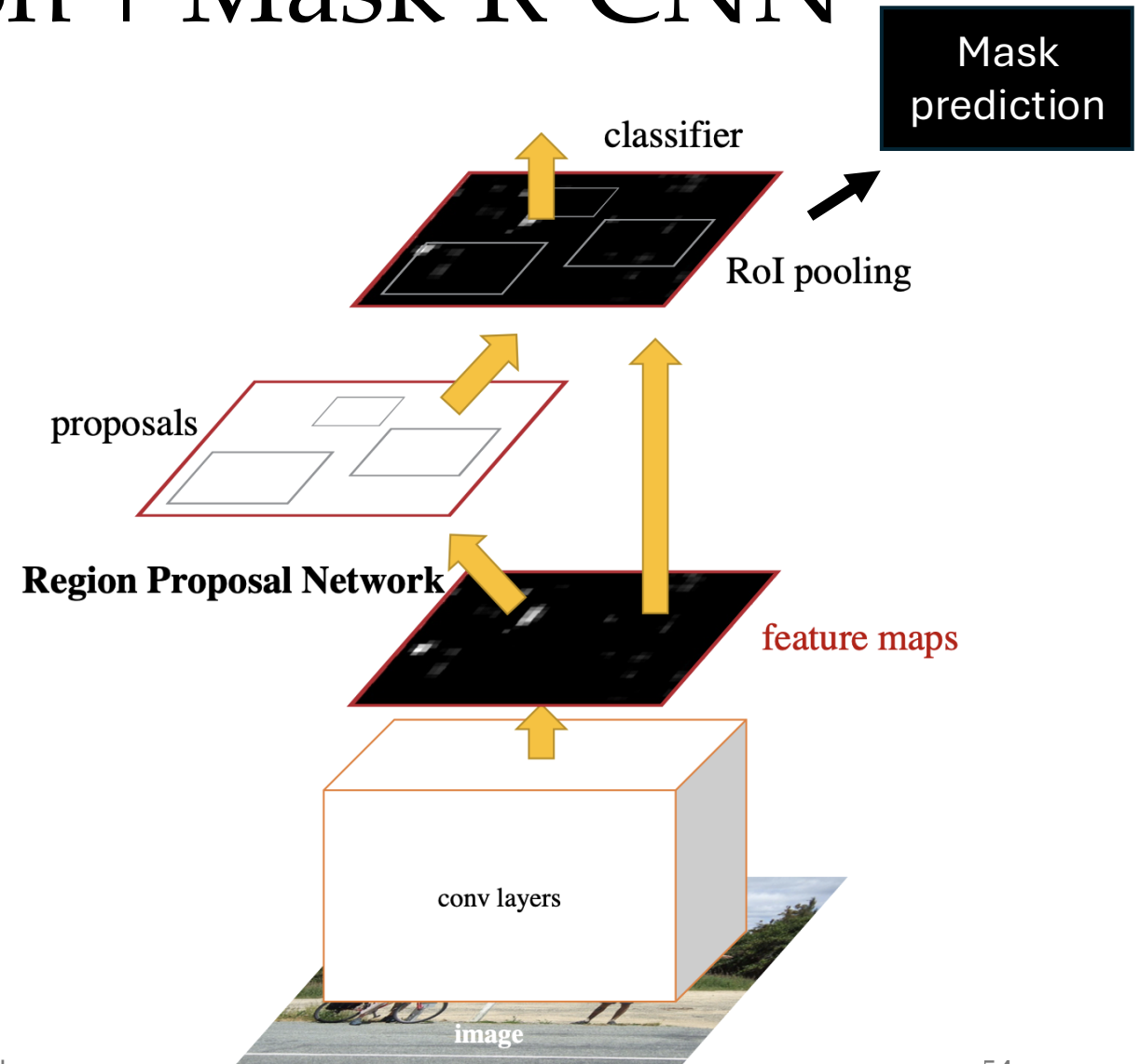
Object detection



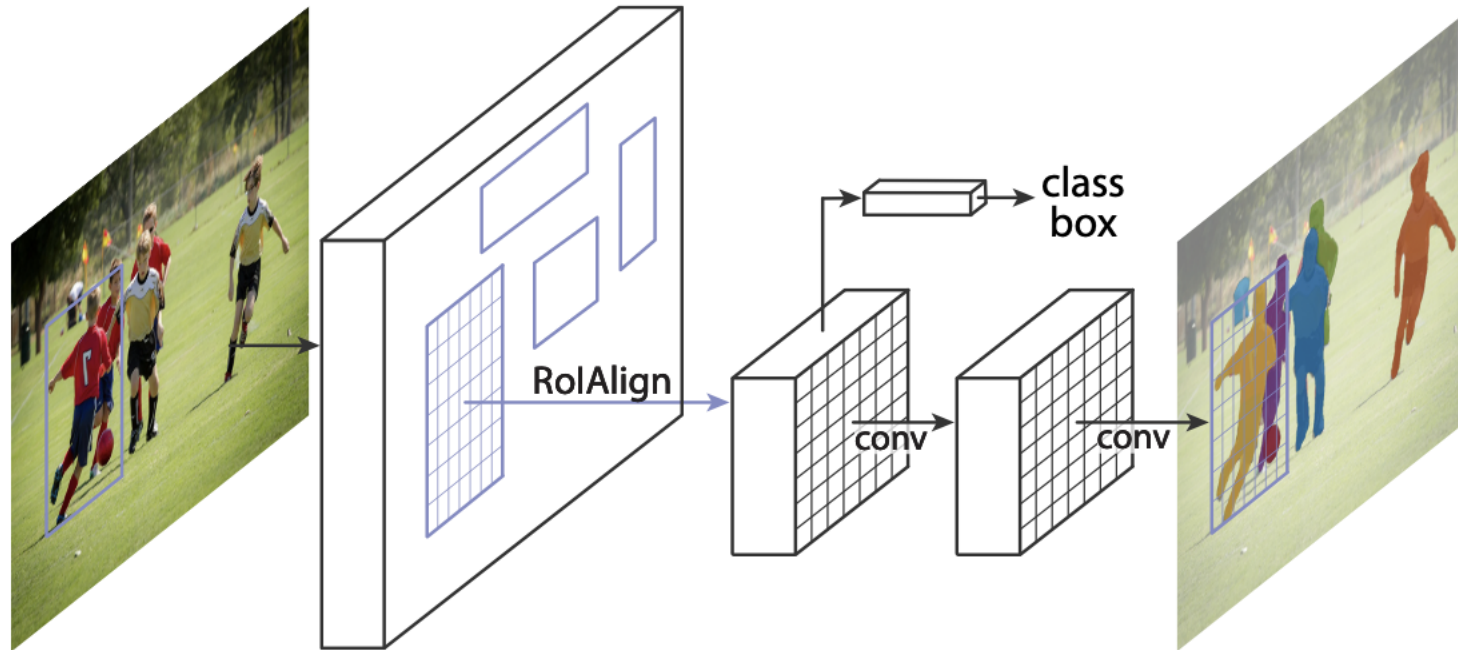
Instance segmentation



Add a small mask network that operates on each RoI and predicts a 28 x 28 binary mask for object presence

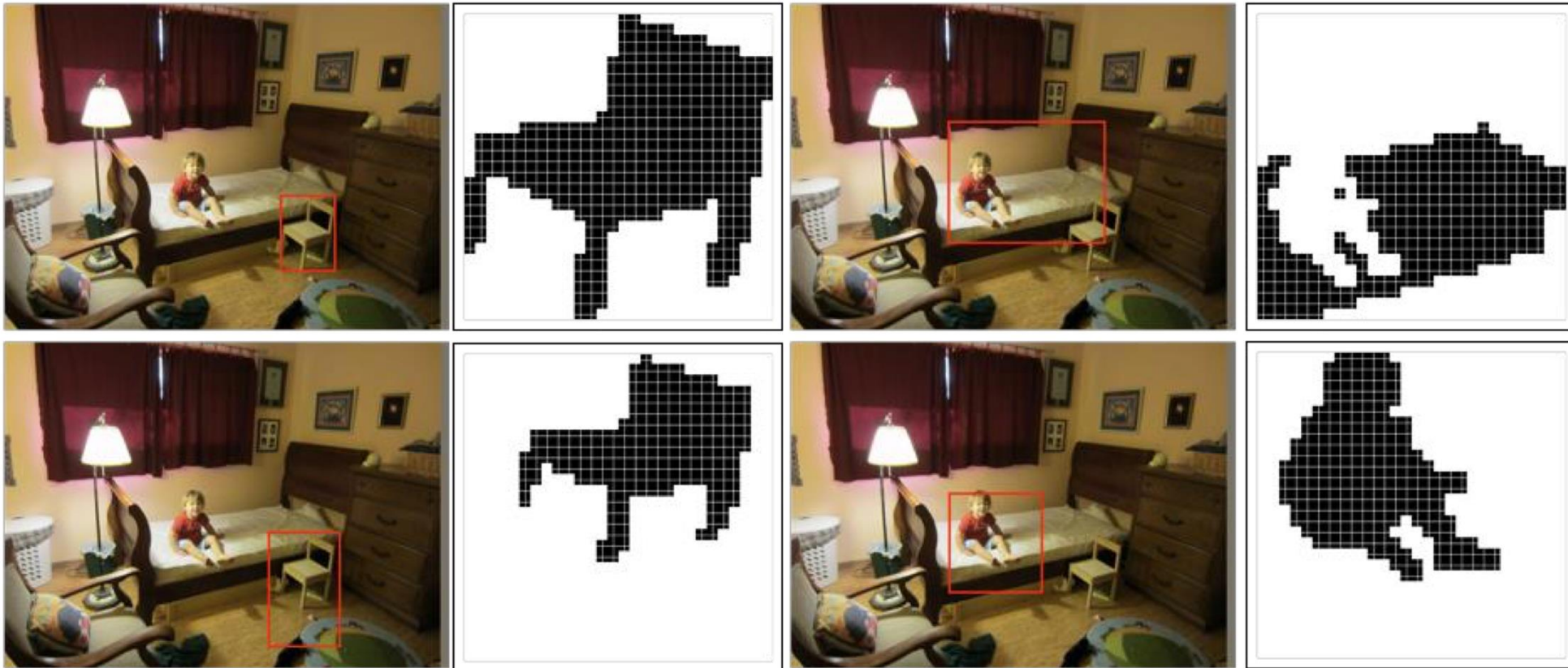


Instance segmentation | Mask R-CNN

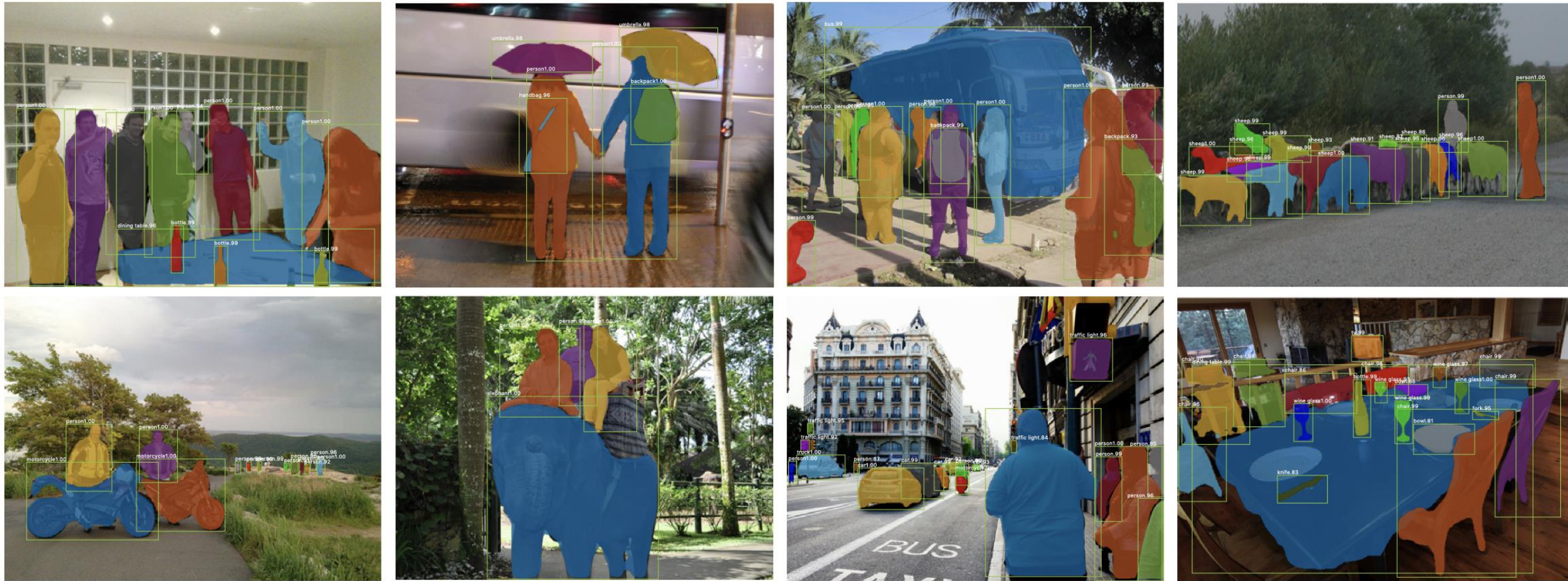


Instance segmentation | Mask R-CNN

Examples of training masks:

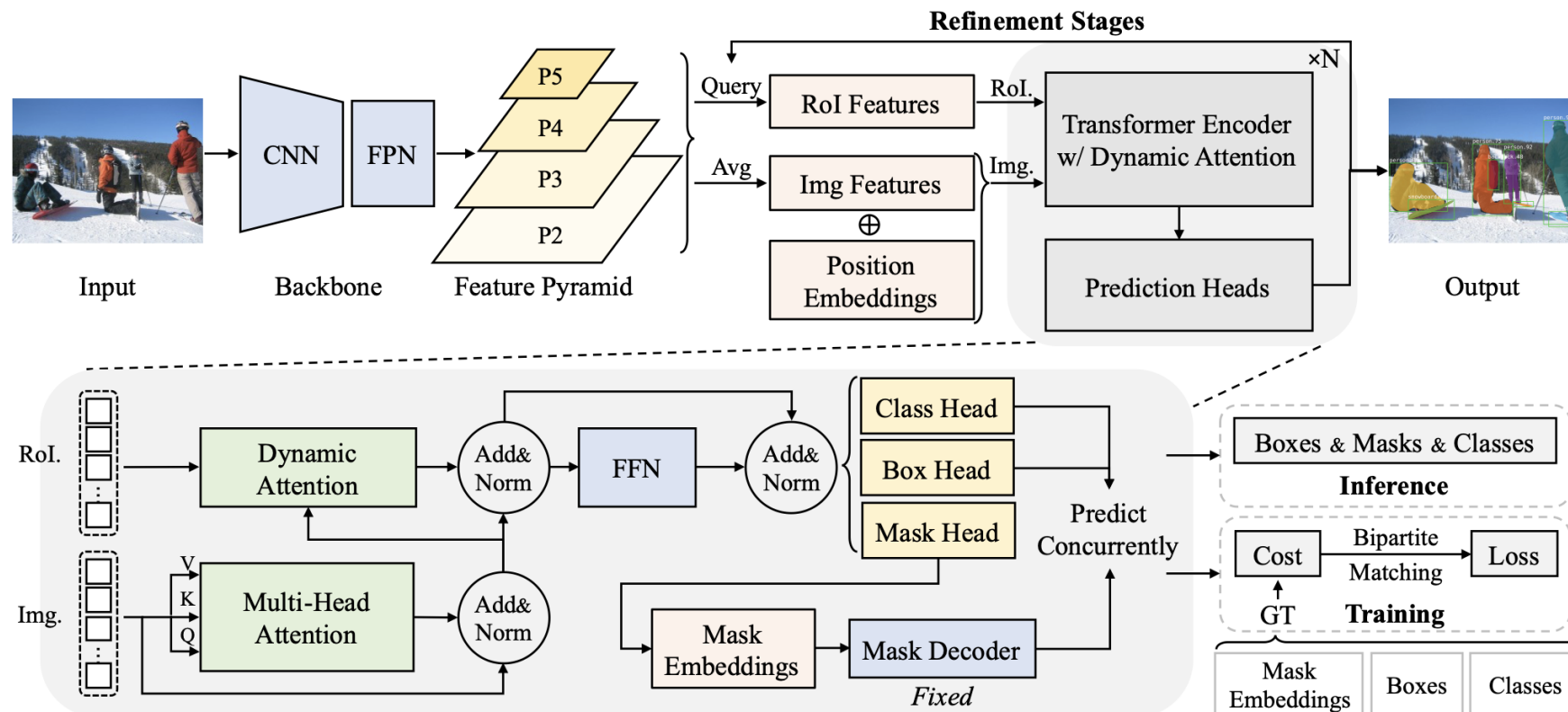


Instance segmentation | Mask R-CNN



Instance segmentation

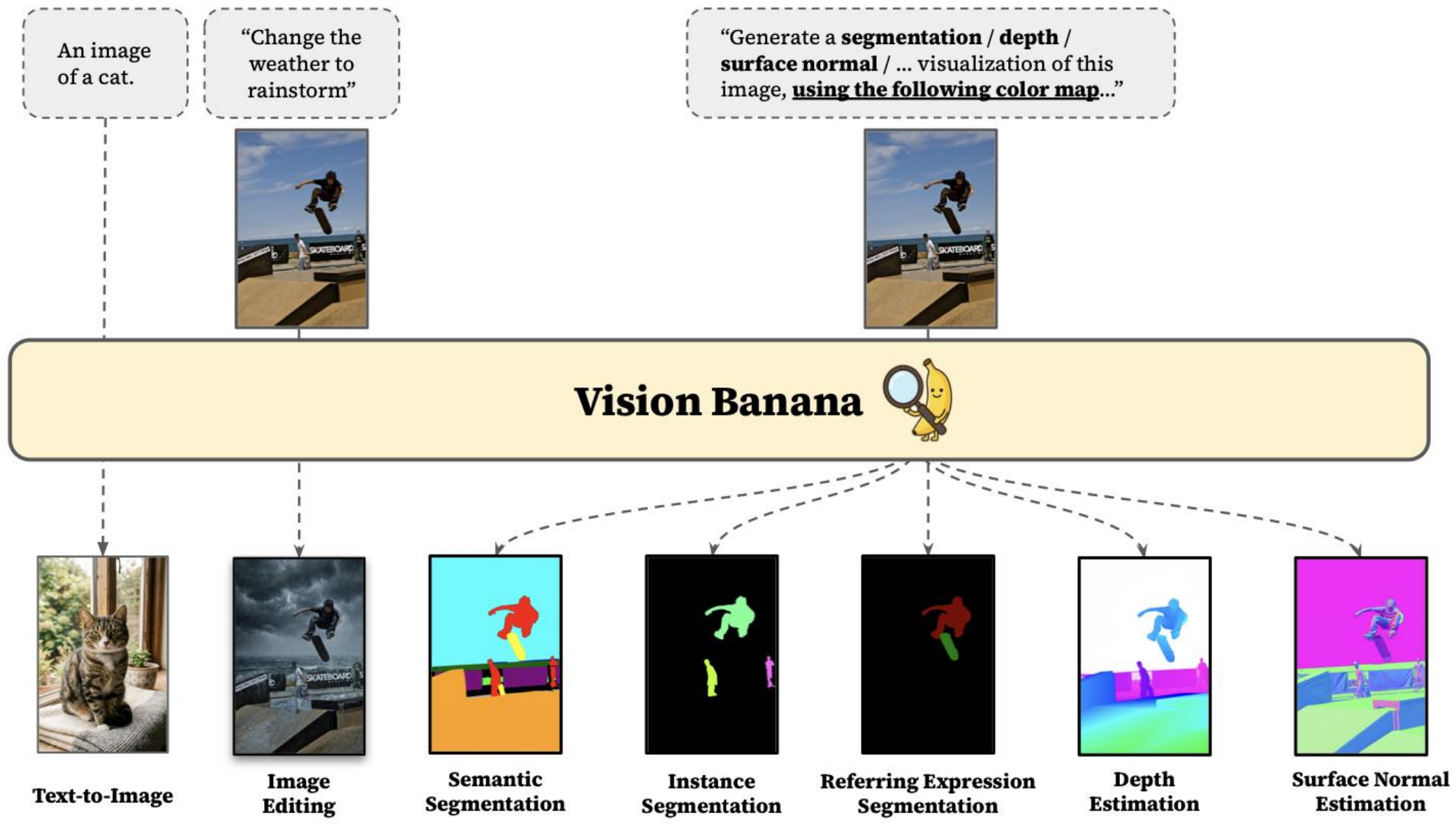
Can be applied to any detection algorithm.



Generative models as multitask models

These are image generation tasks!

Can general generative models solve them?



Computer vision tasks

Image classification



Cat

No spatial context

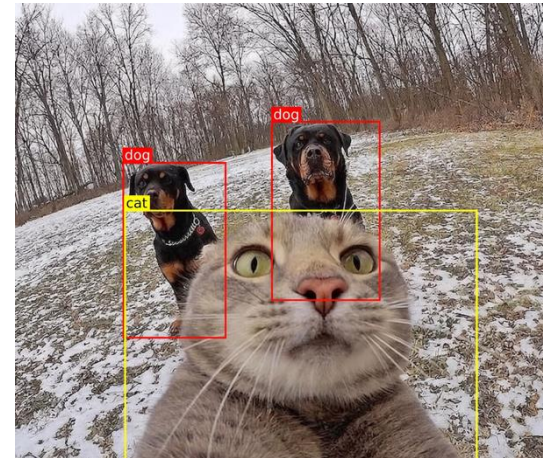
Semantic segmentation



Cat, Dog, Grass, Trees

Multiple objects,
No instance

Object detection



Cat, Dog

Multiple object,
Multiple instances

Instance segmentation



Cat, Dog, Dog